

18/1/18

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UNIT-1

Electromagnetic Theory

$$\text{Del operator} = \nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

$$\phi = 3xy$$

(scalar)

$$\text{grad. } \phi = \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot 3xy$$

(gradient)

$$= \frac{\partial(3xy)}{\partial x} \hat{i} + \frac{\partial(3xy)}{\partial y} \hat{j} + \frac{\partial(3xy)}{\partial z} \hat{k}$$

$$= 3y \hat{i} + 3x \hat{j} + 0$$

$$\vec{E} = -\frac{\partial V}{\partial x} \rightarrow \text{gradient of voltage is el. field}$$

$$\text{let } \vec{A} = 3x^2y \hat{i} + 2xyz \hat{j} + 4y^2z \hat{k}$$

$$\text{div. } \vec{A} = \vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (3x^2y \hat{i} + 2xyz \hat{j} + 4y^2z \hat{k})$$

$$= \frac{\partial(3x^2y)}{\partial x} + \frac{\partial(2xyz)}{\partial y} + \frac{\partial(4y^2z)}{\partial z} \rightarrow \text{dot product}$$

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (3x^2y \hat{i} + 2xyz \hat{j} + 4y^2z \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2y & 2xyz & 4y^2z \end{vmatrix}$$

→ cross product

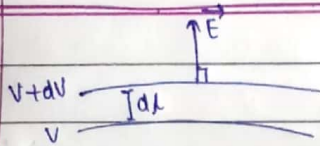
Physical significance of gradient:

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$= \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$= \nabla \phi \cdot d\vec{l} = 0$$

(when $\theta = 90^\circ$)



$$\text{let } \phi = V$$

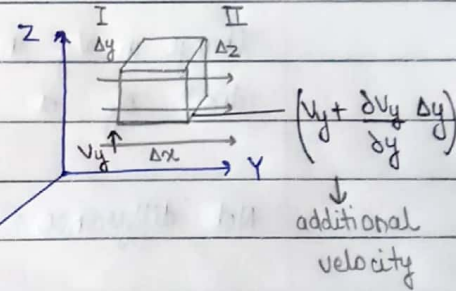
$$d\phi = dV = \nabla V \cdot d\vec{l} = 0$$

∠ b/w them is 90°

So, gradient explains the direction of field.

Physical significance of divergence:

Physical interpretation of divergence is net outward flow per unit volume.



$$\text{mass of liquid entering in I face per unit time} = \rho v_y \Delta x \Delta z$$

volume

density of liquid $\rightarrow \rho$

$$\text{mass of liquid exiting from II face per unit time} = \rho \left(v_y + \frac{\partial v_y}{\partial y} \Delta y \right) \Delta x \Delta z$$

$$\text{Change in net outward flow in y-direction} = \rho \frac{\partial v_y}{\partial y} \Delta x \Delta y \Delta z = \frac{\partial (\rho v_y)}{\partial y} \Delta x \Delta y \Delta z$$

$$\text{Similarly, change in net outward flow in x-direction} = \frac{\partial (\rho v_x)}{\partial x} \Delta x \Delta y \Delta z$$

$$\text{Change in net outward flow in z-direction} = \frac{\partial (\rho v_z)}{\partial z} \Delta x \Delta y \Delta z$$

$$\text{Net outward flow per unit volume} = \frac{\partial (\rho v_x)}{\partial x} + \frac{\partial (\rho v_y)}{\partial y} + \frac{\partial (\rho v_z)}{\partial z} = \nabla \cdot (\rho \mathbf{v}) \quad \left[\begin{array}{l} \text{can be 0} \\ \text{or non-zero} \end{array} \right]$$

(divided by $\Delta x \Delta y \Delta z$)

• if $\nabla \cdot \rho \mathbf{v} = 0 \rightarrow$ flow of liquid is uniform

This is called as SOLENOIDAL condition.

if $\nabla \cdot \rho \mathbf{v} \neq 0 \rightarrow$ flow of liquid is non-uniform.

This is called as NON-SOLENOIDAL condition.

Also, if $\nabla \cdot \rho \mathbf{v} \rightarrow +ve$ Compressive (more pressure afterwards) eg. boiler

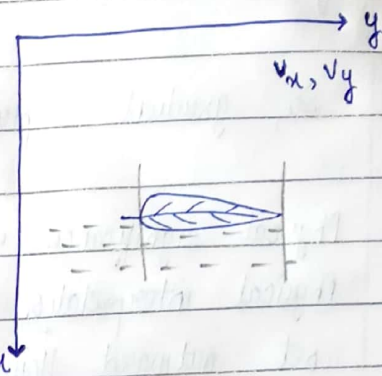
if $\nabla \cdot \rho \mathbf{v} \rightarrow -ve$ Expansion eg. balloon

Physical significance of curl:

It defines the rotational motion.

Change in v_x in the dis.ⁿ of v_y = $v_x + \frac{\partial v_x}{\partial y} dy$
(in clockwise dis.ⁿ)

Change in v_y in the dis.ⁿ of x = $v_y + \frac{\partial v_y}{\partial x} dx$
(in anti-clockwise dis.ⁿ)



$$\text{Net difference} = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = (\nabla \times \mathbf{v})_z$$

motion is in
z dis.ⁿ

$$(\nabla \times \mathbf{v})_x = \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}$$

$$(\nabla \times \mathbf{v})_y = \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}$$

$\hat{i}$	$\hat{j}$	$\hat{k}$
$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial}{\partial z}$
v_x	v_y	v_z

$$\text{Net rotation} = (\nabla \times \mathbf{v}) = () \hat{i} + () \hat{j} + () \hat{k}$$

- if $\nabla \times \mathbf{v} = 0 \rightarrow$ No rotation; linear motion IRROTATIONAL motion
- if $\nabla \times \mathbf{v} \neq 0 \rightarrow$ Rotates ROTATIONAL motion

★ Divergence Theorem: The volume integral of the divergence of vector field $\vec{A}$ taken on any volume V is equal to surface integral of $\vec{A}$ taken over the closed surface that bounds volume V .

$$\int_V (\nabla \cdot \vec{A}) dV = \oint_S \vec{A} \cdot d\vec{S}$$

$\downarrow$ $\downarrow$
 volume integral surface integral
 (surface should be closed)

Stoke's Theorem: The surface integral of curl of vector field $\vec{A}$ over an open surface equals line integral of vector field $\vec{A}$ over the closed curve bounding the surface area.

$$\int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} = \oint \vec{A} \cdot d\vec{l}$$

$\downarrow$ dot product $\downarrow$ dot product

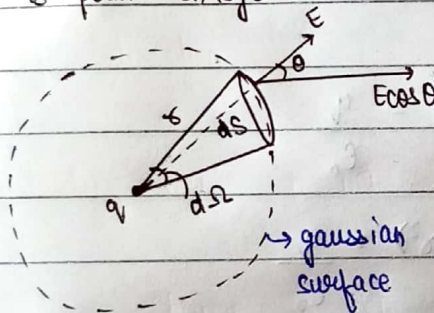
Gauss Theorem

The flux of the electric field $\vec{E}$ through any closed surface equals the total net charge inside the enclosed volume divided by ϵ_0 .

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

$\epsilon_0 \rightarrow$ electric permittivity in vacuum

q is point charge.



Flux due to dS , $d\phi = E \cos \theta dS$

We know, $E = \frac{q}{4\pi\epsilon_0 r^2}$

$$d\phi = \frac{q}{4\pi\epsilon_0 r^2} \cos \theta dS$$

$d\Omega \rightarrow$ solid angle

$$\text{or } d\phi = \frac{q}{4\pi\epsilon_0} \times \frac{\cos \theta dS}{r^2}$$

Integrating, $\int d\phi = \frac{q}{4\pi\epsilon_0} \int \frac{\cos \theta dS}{r^2}$

$$\frac{\cos \theta dS}{r^2} = d\Omega = 4\pi \quad (\text{for sphere})$$

$$\text{So, } \phi = \frac{q}{4\pi\epsilon_0} \cdot 4\pi$$

$$\boxed{\phi = \frac{q}{\epsilon_0}}$$

$$\phi = \int E \cos \theta \, dS = \frac{q}{\epsilon_0}$$

$$\boxed{\phi = \int \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}}$$

NOTE: If many point charges are present, then

$$\phi = \int \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \sum_{i=1}^n q_i$$

Applying Divergence Theorem,

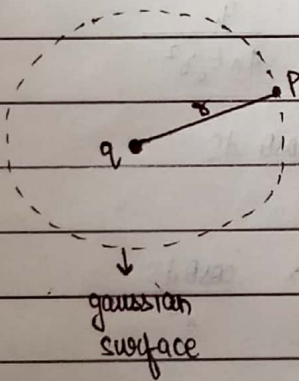
$$\int_V (\vec{\nabla} \cdot \vec{E}) \, dv = \frac{q}{\epsilon_0}$$

- If charge is spread over a volume having charge density ρ ,

$$\boxed{\int_V (\vec{\nabla} \cdot \vec{E}) \, dv = \frac{q}{\epsilon_0} = \frac{1}{\epsilon_0} \int \rho \, dv} \rightarrow \text{Gauss Divergence Theorem}$$

Applications:

- Field due to a point charge:



$$\int \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

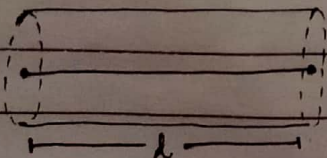
$$E \times (4\pi r^2) = \frac{q}{\epsilon_0}$$

$$\boxed{E = \frac{q}{4\pi\epsilon_0 r^2}}$$

If another point charge q' is present at P,
force b/w 2 charges, $F = q'E$
(q and q')

$$F = \frac{qq'}{4\pi\epsilon_0 r^2}$$

- Field due to a line charge:



$\lambda \rightarrow$ charge density per unit length
So, $q = \lambda l$

$$\int \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$E \times 2\pi r l = \frac{\lambda l}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

3. Field due to an infinite plane surface (parallel plate):

Charge q is distributed all over the plate.

$A \rightarrow$ surface area of plate

$\sigma \rightarrow$ charge density per unit area



$$\int \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

considering both sides $\swarrow$

$$(E \times A + E \times A) = \frac{\sigma A}{\epsilon_0}$$

$$2EA = \frac{\sigma A}{\epsilon_0}$$

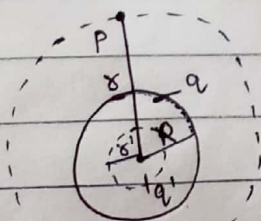
$$E = \frac{\sigma}{2\epsilon_0}$$

• If there are 2 parallel plates,

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

4. Field due to a sphere:



$\hookrightarrow$ gaussian surface

• Outside the sphere, $\int \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$

$$E \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

- Inside the sphere, $E \times 4\pi r^2 = \frac{q'}{\epsilon_0}$

$$E = \frac{q'}{4\pi\epsilon_0 r^2}$$

where, $q' = \frac{q}{\frac{4}{3}\pi R^3} \times \frac{4}{3}\pi r^3$

$$q' = \frac{qr^3}{R^3}$$

- If charge is present on the surface not inside, $\vec{E} = 0$.

Q. Given the potential field, $V = 2x^2y - 5z$, and point P (-4, 3, 6), find out

(a) potential V (b) electric field intensity at point P

Sol.ⁿ (b) $E = -\nabla V = -4xy\hat{i} - 2x^2\hat{j} + 5\hat{k} = 48\hat{i} - 32\hat{j} + 5\hat{k}$

$$|\vec{E}| = [(48)^2 + (-32)^2 + (5)^2]^{1/2} = 57.9$$

(a) $V = 2x^2y - 5z = 96 - 30 = 66 \text{ V}$

Q. Find gradient of a scalar function of position v , where $v(x, y, z) = x^2y + e^z$. Calculate the magnitude of gradient at point P (1, 5, -2).

Sol.ⁿ $\vec{v} = \nabla(v) = \frac{\partial (x^2y + e^z)}{\partial x}\hat{i} + \frac{\partial (x^2y + e^z)}{\partial y}\hat{j} + \frac{\partial (x^2y + e^z)}{\partial z}\hat{k}$

$$= (2xy)\hat{i} + x^2\hat{j} + e^z\hat{k}$$

$$= 10\hat{i} + \hat{j} + e^{-2}\hat{k}$$

$$|\vec{v}| = \sqrt{100 + 1 + e^{-4}} = \sqrt{101 + e^{-4}}$$

Q. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that

(a) $\vec{\nabla} \cdot \vec{r} = 3$ (b) $\vec{\nabla} \times \vec{r} = 0$

Sol.ⁿ (a) $\vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$

$$\vec{\nabla} \cdot \vec{r} = \frac{\partial (x)}{\partial x} + \frac{\partial (y)}{\partial y} + \frac{\partial (z)}{\partial z} = 1 + 1 + 1 = 3$$

$$(b) \quad \vec{\nabla} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \hat{i} \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \right) - \hat{j} \left(\frac{\partial z}{\partial x} - \frac{\partial x}{\partial z} \right) + \hat{k} \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right)$$

$$= 0$$

Q. If $\vec{a} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$, show that the system is either rotational or solenoidal at $(1, -1, 1)$.

Sol.ⁿ $\vec{\nabla} \cdot \vec{a} = \frac{\partial}{\partial x}(xz^3) + \frac{\partial}{\partial y}(-2x^2yz) + \frac{\partial}{\partial z}(2yz^4)$

$$= z^3 - 2x^2z + 8yz^3$$

$$= 1 - 2 - 8 = -9 \neq 0 \Rightarrow \text{Non-solenoidal}$$

$$\vec{\nabla} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(2yz^4) + \frac{\partial}{\partial z}(2x^2yz) \right] - \hat{j} \left[\frac{\partial}{\partial x}(2yz^4) - \frac{\partial}{\partial z}(xz^3) \right] + \hat{k} \left[\frac{\partial}{\partial x}(-2x^2yz) - \frac{\partial}{\partial y}(xz^3) \right]$$

$$= \hat{i}(2z^4 + 2x^2y) - \hat{j}(-3xz^2) + \hat{k}(-4xyz)$$

$$= (2z^4 + 2x^2y)\hat{i} + (3xz^2)\hat{j} - (4xyz)\hat{k}$$

$$= 4\hat{i} + 3\hat{j} + 4\hat{k} \neq 0 \Rightarrow \text{Rotational}$$

Q. Find the value of constant 'c' for which $\vec{A} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+cz)\hat{k}$ is solenoidal.

Sol.ⁿ $\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+cz)$

$$= 1 + 1 + c = c + 2$$

For $\vec{\nabla} \cdot \vec{A}$ to be solenoidal, $\vec{\nabla} \cdot \vec{A} = 0$ so, $c + 2 = 0$

$$\boxed{c = -2}$$

Q. Show that $\oint_S \vec{F} \cdot d\vec{s} = 6V$, where S is a closed surface enclosing a

volume V and $\vec{F} = 2x\hat{i} + 3y\hat{j} + z\hat{k}$.

Sol.ⁿ Applying Divergence theorem, $\int_V (\vec{\nabla} \cdot \vec{A}) dV = \oint_S \vec{A} \cdot d\vec{s}$

$$\oint_S \vec{F} \cdot d\vec{s} = \int_V \vec{\nabla} \cdot \vec{F} dV = \int_V \left(\frac{d}{dx}(2x) + \frac{d}{dy}(3y) + \frac{d}{dz}(z) \right) dV$$

$$= \int_V (2+3+1) dV = 6 \int_V dV$$

$$= 6V \quad \text{Hence, proved.}$$

Q. Evaluate both sides of Stoke's theorem for field $\vec{H} = 6xy\hat{i} - 3y^2\hat{j}$ A/m and rectangular path around region, $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z=0$. let the +ve dir.ⁿ of $d\vec{s}$ be $\hat{k}$.

Sol.ⁿ $\oint_C \vec{H} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{H}) \cdot d\vec{s}$

$$\text{RHS} = \vec{\nabla} \times \vec{H} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{vmatrix} = -6x\hat{k}$$

$$\text{RHS} = \int_S (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \int_S -6x\hat{k} (\hat{k} \cdot d\vec{s}) = - \int_S 6x dxdy$$

$$= - \int_{y=-1}^1 \int_{x=2}^5 6x dxdy$$

$$= -6 \left[\frac{x^2}{2} \right]_2^5 \left[dy \right]_{-1}^1 = -6 \times \frac{1}{2} \times (25-4) \times 2$$

$$= -6 \times 21 = -126$$

$$\text{LHS} = \oint_C \vec{H} \cdot d\vec{l} = \oint_C (6xy\hat{i} - 3y^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \oint_C (6xydx - 3y^2dy)$$

$$\oint_C \vec{H} \cdot d\vec{l} = \int_{x=5, y=1}^2 \int 6xy dx + \int_{x=2, y=1}^5 \int 6xy dx$$

$$= \frac{6x^2}{2} \Big|_5^2 + \frac{6x^2}{2} \Big|_2^5 (-1) = \frac{[x^2]^2}{5} + (-1) \frac{[x^2]^5}{2} = -63 - 63 = -126 = \text{RHS}$$

Ampere's Law

Maxwell's equation, $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$

$\vec{B} \rightarrow$ magnetic flux density

$\vec{J} \rightarrow$ current density

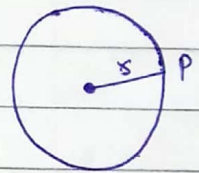
Integrating, $\int (\vec{\nabla} \times \vec{B}) \cdot d\vec{s} = \mu_0 \int \vec{J} \cdot d\vec{s}$

$$\int (\vec{\nabla} \times \vec{B}) \cdot d\vec{s} = \mu_0 I$$

Using Stoke's Theorem, $\int (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} = \oint \vec{A} \cdot d\vec{l}$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

for a circular enclosed surface of radius r ,



$$B \times 2\pi r = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$\mu_0 \rightarrow$ magnetic permeability $= 4\pi \times 10^{-7}$

field at a point inside a long cylindrical wire,

$$\oint B \cdot dl = \mu_0 I$$

$$B \times 2\pi R = \mu_0 I$$

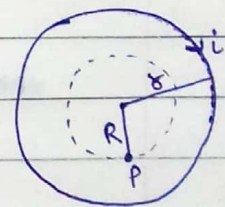
$$B = \frac{\mu_0 I}{2\pi R}$$

$I \rightarrow$ current enclosed by circle of radius R

$$I = \frac{i}{\pi r^2} \cdot \pi R^2 = \frac{i R^2}{r^2}$$

So,

$$B = \frac{\mu_0 i R}{2\pi r^2}$$



Magnetic flux density within a long solenoid,
 $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

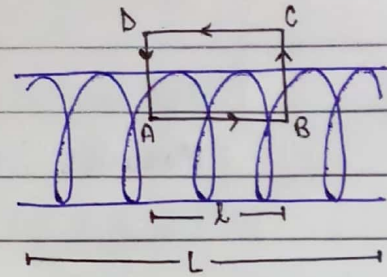
$$\Rightarrow \int_A^B \vec{B} \cdot d\vec{l} + \int_{B_0}^C \vec{B} \cdot d\vec{l} + \int_{C_0}^D \vec{B} \cdot d\vec{l} + \int_{D_0}^A \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\Rightarrow \int_A^B \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$Bl = \mu_0 I$$

$$I = i \left(\frac{N}{L} \right) l$$

So, $B = \frac{\mu_0 i N}{L}$



$N \rightarrow$ no. of turns

$i \rightarrow$ total current

$I \rightarrow$ current through AB

Maxwell's Equations

	Differential form		Integral form
1.	$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ (Gauss law)	$\rightarrow$	$\oint \vec{E} \cdot d\vec{s} = \int \rho dv$
2.	$\vec{\nabla} \cdot \vec{B} = 0$ (Eq. ⁿ for monopoles)	$\rightarrow$	$\oint \vec{B} \cdot d\vec{s} = 0$
3.	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ (Faraday's law)	$\rightarrow$	$\oint \vec{E} \cdot d\vec{l} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$
4.	$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{J}$ (Ampere's law)	$\rightarrow$	$\oint \vec{H} \cdot d\vec{l} = I + \int \frac{\partial \vec{D}}{\partial t} \cdot d\vec{s}$

where

$D \rightarrow$ electric displacement ; $D = \epsilon E$

$\sigma \rightarrow$ electrical conductivity ; $J = \sigma E$

$B \rightarrow$ magnetic flux density ; $B = \mu H$

$J \rightarrow$ current density

$H \rightarrow$ magnetic field intensity

$c \rightarrow$ velocity of light ; $c = \frac{1}{\sqrt{\mu \epsilon}}$

Proofs:

1. $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

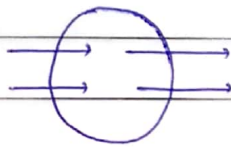
Using Gauss law, $\int \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$

$$\int (\vec{\nabla} \cdot \vec{E}) dv = \frac{1}{\epsilon_0} \int \rho dv$$

$$\int (\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0}) dv = 0$$

Since dv can't be 0 so, $\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0} = 0 \Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

2. $\vec{\nabla} \cdot \vec{B} = 0$



N S

Poles can't be separated

No. of lines entering and leaving the surface is same.

So, Net flux = 0

$$\therefore \phi = \int \vec{B} \cdot d\vec{s} = 0$$

By using divergence theorem, $\int (\vec{\nabla} \cdot \vec{B}) dv = 0$

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

3. $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Induction due to magnetic field, $e = -\frac{\partial \phi}{\partial t}$

$$\phi = \int \vec{B} \cdot d\vec{s}$$

$$e = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

Induction due to electric field, $e = \int \vec{E} \cdot d\vec{l}$

$$\text{So, } \int \vec{E} \cdot d\vec{l} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

Applying Stoke's Theorem,

$$\int (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$\int \left[(\vec{\nabla} \times \vec{E}) + \frac{\partial \vec{B}}{\partial t} \right] \cdot d\vec{s} = 0$$

Since $d\vec{s}$ can't be 0, so $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

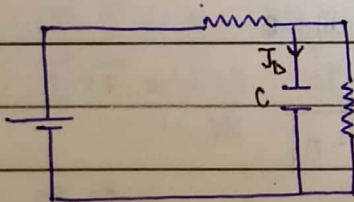
4. $\vec{\nabla} \times \vec{B} - \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} = \mu \vec{J}$

$$\int \vec{B} \cdot d\vec{l} = \mu I \quad (\text{Ampere's law})$$

Applying Stoke's theorem, $\int (\vec{\nabla} \times \vec{B}) \cdot d\vec{s} = \mu \int \vec{J} \cdot d\vec{s}$

$$\int (\vec{\nabla} \times \vec{B} - \mu \vec{J}) \cdot d\vec{s} = 0$$

Since $d\vec{s}$ can't be 0, so $\vec{\nabla} \times \vec{B} = \mu \vec{J} \rightarrow$ (when capacitor is absent from circuit)



$J_d \rightarrow$ displacement current

$$\text{So, } \vec{\nabla} \times \vec{B} = \mu (\vec{J} + \vec{J}_d)$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad [\because D = \epsilon_0 E]$$

$$\text{So, } \vec{\nabla} \times \vec{B} = \mu \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

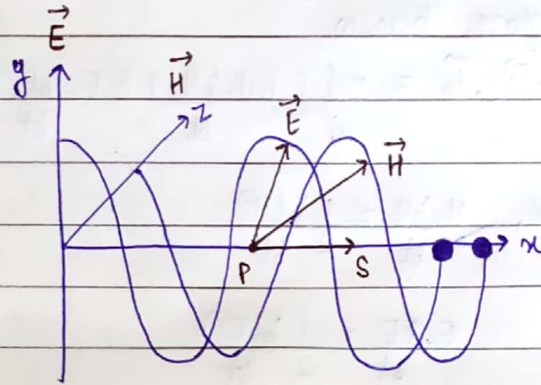
$$\vec{\nabla} \times \vec{B} = \mu \vec{J} + \mu \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\boxed{\vec{\nabla} \times \vec{B} = \mu \vec{J} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}}$$

Poynting Theorem

This is an important aspect of wave propagation and it is the flow of power through space. As em waves propagate through space from source to receiver, there exists a simple and direct relationship b/w power transferred and amplitude of electric and magnetic field strength.



$$\vec{S} = \vec{E} \times \vec{H}$$

$$S = \frac{P}{\text{Area}} = \frac{P}{4\pi r^2}$$

$P \rightarrow$ Power

$S \rightarrow$ Energy

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = -\frac{\partial}{\partial t} \int_V \left[\frac{\mu H^2}{2} + \frac{\epsilon E^2}{2} \right] dV - \int_V (\vec{E} \cdot \vec{J}) dV$$

Rate of
energy flow

Rate of decrease
of stored energy

Total instantaneous ohmic power
dissipated within volume

$$\vec{\nabla} \cdot \vec{D} = \rho$$

or

$$\vec{\nabla} \cdot \vec{D} = \rho$$

-(i)

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

-(ii)

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

-(iii)

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

-(iv)

Proof:

Take dot product of eq. (iii) with $\vec{H}$,

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t}$$

Multiply (dot product) by $\vec{E}$, i.e. eq. (iv), $\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} - \vec{E} \cdot \vec{J}$$

Take volume integral on both sides,

$$\int_V [\vec{\nabla} \cdot (\vec{E} \times \vec{H})] dV = - \int_V \left[\mu \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} + \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right] dV - \int_V (\vec{E} \cdot \vec{J}) dV$$

Applying Divergence theorem,

$$\int_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = - \int_V \left[\mu \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} + \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right] dV - \int_V (\vec{E} \cdot \vec{J}) dV$$

We know, $\vec{H} \cdot \frac{\partial \vec{H}}{\partial t} = \frac{1}{2} \frac{\partial (H^2)}{\partial t}$

$$\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} = \frac{1}{2} \frac{\partial (E^2)}{\partial t}$$

$$\text{So, } \int_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = - \int_V \left[\frac{\mu}{2} \frac{\partial H^2}{\partial t} + \frac{\epsilon}{2} \frac{\partial E^2}{\partial t} \right] dV - \int_V (\vec{E} \cdot \vec{J}) dV$$

$$\int_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = - \frac{\partial}{\partial t} \int_V \left[\frac{\mu}{2} H^2 + \frac{\epsilon}{2} E^2 \right] dV - \int_V (\vec{E} \cdot \vec{J}) dV$$

Hence, proved.

Electromagnetic Wave Equation

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} + \mu \vec{J}$$

- Q. A uniformly charged hollow sphere has a total charge of $300 \mu\text{C}$ and a radius of 8 cm . Find the electric intensity at
- a point 16 cm from the centre of the sphere
 - at a point on the surface of the sphere
 - at a point 4 cm from the centre of sphere

Sol.ⁿ (i) $E = \frac{kq}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} = \frac{9 \times 10^9 \times 300 \times 10^{-6}}{(16)^2 \times 10^{-4}}$

$$E = 4.22 \times 10^8 \text{ V/m}$$

(ii) $E = \frac{9 \times 10^9 \times 300 \times 10^{-6}}{64 \times 10^{-4}} = 1.05 \times 10^8 \text{ V/m}$

(iii) $E = \frac{kq}{R^3} = \frac{9 \times 10^9 \times 300 \times 10^{-6} \times r}{R^3} = 0$ (as it is hollow sphere)

- Q. Calculate the electric field intensity due to a spherically charged distribution given by $\rho = \rho_0 \left(1 - \frac{R}{a}\right)$, $R \leq a$, and $\rho = 0$ for $R > a$ at points inside and outside.

Sol.ⁿ

$$dV = 4\pi R^2 dR$$

$a \rightarrow$ radius of sphere

Applying Divergence theorem, $\int \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int \rho dV$

inside,

$$\int E (4\pi R^2) = \frac{1}{\epsilon_0} \int_0^R \rho_0 \left(1 - \frac{R}{a}\right) 4\pi R^2 dR$$

$$\int E (4\pi R^2) = \frac{\rho_0 \cdot 4\pi}{\epsilon_0} \int_0^R \left(R^2 - \frac{R^3}{a}\right) dR = \frac{4\pi \rho_0}{\epsilon_0} \left[\frac{R^3}{3} - \frac{R^4}{4a} \right]$$

$$E (4\pi R^2) = \frac{4\pi \rho_0}{\epsilon_0} \left[\frac{R}{3} - \frac{R^2}{4a} \right] R^2$$

$$E = \frac{\rho_0}{\epsilon_0} \left[\frac{R}{3} - \frac{R^2}{4a} \right]$$

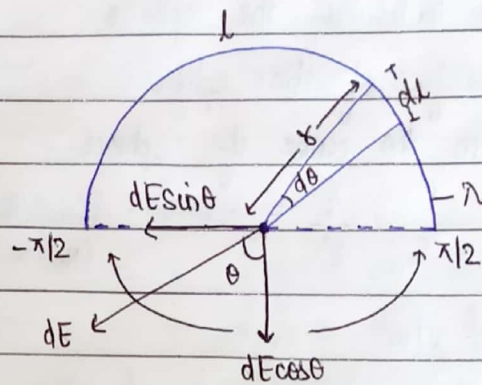
outside,

$$\int E (4\pi R^2) = \frac{1}{\epsilon_0} \int_0^a \rho_0 \left(1 - \frac{R}{a}\right) 4\pi R^2 dR = \frac{\rho_0 \cdot a^3 \cdot 4\pi}{\epsilon_0 \cdot 12}$$

$$E = \frac{\rho_0 a^3}{12 \epsilon_0 R^2}$$

- Q. Find the electric field at the centre of a uniformly charged semi-circular arc.

Sol.ⁿ



Horizontal components ($dE \sin \theta$) will cancel out each other.

$$dE = \frac{\lambda dl \cos \theta}{4\pi\epsilon_0 r^2}$$

Integrating both sides

$$\int_{\cos \theta} dE = \int_{-\pi/2}^{\pi/2} \frac{\lambda dl \cos \theta}{4\pi\epsilon_0 r^2}$$

$$dl = r d\theta$$

$$E = \int_{-\pi/2}^{\pi/2} \frac{\lambda r \cos \theta d\theta}{4\pi\epsilon_0 r^2}$$

$$E = \frac{\lambda}{4\pi\epsilon_0 r} \left[\sin \theta \right]_{-\pi/2}^{\pi/2} = \frac{2\lambda}{4\pi\epsilon_0 r}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

- Q. Starting from Maxwell's equations, establish Coulomb's law.

Sol.ⁿ (i) $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

(iii) $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

(ii) $\vec{\nabla} \cdot \vec{B} = 0$

(iv) $\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} + \mu \vec{J}$

Using eq. (i), $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

$$\int (\vec{\nabla} \cdot \vec{E}) dV = \int \frac{\rho}{\epsilon_0} dV$$

Using Stoke's Theorem, $\int \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$

$$E = \frac{q}{4\pi\epsilon_0 R^2}$$

$$F = qE$$

$$F = \frac{qq_1}{4\pi\epsilon_0 R^2}$$

Q. Show that equation of continuity $\text{div. } \vec{J} + \frac{\partial \rho}{\partial t} = 0$ is contained in

Maxwell's equations.

OR

Starting from Maxwell's equations, establish the equation of continuity.

Sol.ⁿ Using (iv) Maxwell eq.ⁿ, $\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$

We know, $B = \mu_0 H$ and $D = \epsilon_0 E$

$$\text{So, } \mu_0 (\vec{\nabla} \times \vec{H}) = \frac{1}{(\mu_0 \epsilon_0)^{-1}} \times \frac{\partial \vec{D}}{\partial t} \cdot \frac{1}{\epsilon_0} + \mu_0 \vec{J}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Taking divergence, $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t}$

$$\vec{\nabla} \cdot \vec{J} = - \frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} = - \frac{\partial \rho}{\partial t}$$

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

$$[\because \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}; \quad \frac{\vec{\nabla} \cdot \vec{D}}{\epsilon_0} = \frac{\rho}{\epsilon_0} \Rightarrow \vec{\nabla} \cdot \vec{D} = \rho]$$

Q. Starting from Maxwell's eq.ⁿ, curl $\vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and curl $\vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$,

show that $\text{div. } \vec{B} = 0$ and $\text{div. } \vec{D} = \rho$.

Sol.ⁿ

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Taking divergence, $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) = -\frac{\partial (\vec{\nabla} \cdot \vec{B})}{\partial t}$

$$\text{So, } \vec{\nabla} \cdot \vec{B} = 0$$

$\vec{\nabla} \cdot \vec{B}$ can't be a constant as magnet can't be separated into monopoles.

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Taking divergence, $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t}$

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} = -\frac{\partial \rho}{\partial t}$$

Eq.ⁿ of continuity, $-\frac{\partial \rho}{\partial t} + \frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} = 0$

$$\frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} = \frac{\partial \rho}{\partial t}$$

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Electromagnetic wave equation

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} + \mu \vec{J}$$

$$\vec{B} = \mu \vec{H}, \quad \vec{D} = \epsilon \vec{E}, \quad \vec{J} = \sigma \vec{E}$$

Considering there is no charge on the medium,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \text{take curl on both sides,}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial (\vec{\nabla} \times \vec{B})}{\partial t}$$

$$\nabla(\vec{\nabla} \cdot \vec{E}) - \nabla^2 E = -\mu \frac{\partial H}{\partial t} = -\mu \frac{\partial}{\partial t} \left[J + \frac{\partial D}{\partial t} \right]$$

(no charge is present)

$$-\nabla^2 E = -\mu \frac{\partial}{\partial t} \left[\sigma E + \epsilon \frac{\partial E}{\partial t} \right]$$

$$-\nabla^2 E = -\mu \sigma \frac{\partial E}{\partial t} - \mu \epsilon \frac{\partial^2 E}{\partial t^2}$$

$$\boxed{\nabla^2 E - \mu \sigma \frac{\partial E}{\partial t} - \mu \epsilon \frac{\partial^2 E}{\partial t^2} = 0}$$

$$\vec{\nabla} \times \vec{H} = \frac{1}{c^2} \left[\vec{J} + \frac{\partial \vec{D}}{\partial t} \right], \text{ take curl on both sides,}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \times \vec{J} + \vec{\nabla} \times \frac{\partial \vec{D}}{\partial t}$$

$$\nabla(\vec{\nabla} \cdot \vec{H}) - \nabla^2 H = \sigma(\vec{\nabla} \times \vec{E}) + \vec{\nabla} \times \frac{\partial \vec{D}}{\partial t}$$

$$-\nabla^2 H = -\sigma \frac{\partial \vec{B}}{\partial t} + \frac{\partial (\vec{\nabla} \times \vec{D})}{\partial t}$$

$$-\nabla^2 H = -\sigma \frac{\partial \vec{B}}{\partial t} + \epsilon \frac{\partial (\vec{\nabla} \times \vec{E})}{\partial t}$$

$$-\nabla^2 H = -\mu \sigma \frac{\partial \vec{H}}{\partial t} + \epsilon \left(-\frac{\partial^2 \vec{B}}{\partial t^2} \right)$$

$$-\nabla^2 H = -\sigma \mu \frac{\partial \vec{H}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

$$\boxed{\nabla^2 H - \sigma \mu \frac{\partial \vec{H}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} = 0}$$

(ii) EM wave equation in free space:

Take $\rho = \sigma = 0$ and $\mu \rightarrow \mu_0$; $\epsilon \rightarrow \epsilon_0$

$$\text{So, } \nabla^2 E - \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} = 0 \quad - (1)$$

$$\nabla^2 H - \mu_0 \epsilon_0 \frac{\partial^2 H}{\partial t^2} = 0 \quad - (2)$$

for SHM, the second derivative D.E. is,

$$\nabla^2 \psi - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0$$

$$\text{or } \frac{\partial^2 \psi}{\partial m^2} - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0$$

Comparing the above eq.ⁿ with eq. (1) and (2),

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s} \quad (\text{speed of light})$$

$$\text{So, } \nabla^2 E - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = 0 \quad \text{and} \quad \nabla^2 H - \frac{1}{c^2} \frac{\partial^2 H}{\partial t^2} = 0$$

$$\Rightarrow E(x,t) = E_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$\Rightarrow H(x,t) = H_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$\text{where } \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{k} = k_x\hat{i} + k_y\hat{j} + k_z\hat{k} = k\hat{n} = \frac{2\pi}{\lambda} \hat{n} = \frac{2\pi\nu}{c} \hat{n}$$

$$\boxed{\vec{k} = \frac{\omega}{c} \hat{n}}$$

$\vec{r} \rightarrow$ position vectors ; $k \rightarrow$ wave no.

Using I Maxwell eq.ⁿ, $\vec{\nabla} \cdot \vec{E} = 0$ (no charge is present)

$$\vec{\nabla} \cdot \vec{E} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot E_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot [E_{0x}\hat{i} + E_{0y}\hat{j} + E_{0z}\hat{k}] e^{i(k_x x + k_y y + k_z z)} e^{i(k_x x + k_y y + k_z z)} e^{i(k_x x + k_y y + k_z z)} e^{-i\omega t}$$

$$= E_{0x} \frac{\partial}{\partial x} e^{i(k_x x + k_y y + k_z z) - i\omega t} + E_{0y} \frac{\partial}{\partial y} e^{i(k_x x + k_y y + k_z z) - i\omega t} +$$

$$E_{0z} \frac{\partial}{\partial z} e^{i(k_x x + k_y y + k_z z) - i\omega t}$$

$$= E_{0x} \times i k_x e^{i(k_x x + k_y y + k_z z) - i\omega t} + E_{0y} \times i k_y e^{i(k_x x + k_y y + k_z z) - i\omega t} +$$

$$E_{0z} \times i k_z e^{i(k_x x + k_y y + k_z z) - i\omega t}$$

$$= i [E_{0x} k_x + E_{0y} k_y + E_{0z} k_z] e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$= i [E_{0x} \hat{i} + E_{0y} \hat{j} + E_{0z} \hat{k}] e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$= i E_0 e^{i\vec{k} \cdot \vec{r} - i\omega t} \cdot \vec{k}$$

$$\boxed{\vec{\nabla} \cdot \vec{E} = i \vec{E} \cdot \vec{k}}$$

Similarly, $\boxed{\vec{\nabla} \cdot \vec{H} = i \vec{H} \cdot \vec{k}}$

Since $\vec{\nabla} \cdot \vec{E} = \vec{\nabla} \cdot \vec{H} = 0$, $i \vec{E} \cdot \vec{k} = i \vec{H} \cdot \vec{k} = 0$

So, $\boxed{\vec{E} \cdot \vec{k} = \vec{H} \cdot \vec{k} = 0}$

↓
dir.ⁿ of wave propagation

Conclusion: This implies the b/w dir.ⁿ of wave propagation and $\vec{E}$ or $\vec{H}$ is 90° so, both $\vec{E}$ and $\vec{H}$ are perpendicular to the dir.ⁿ of wave propagation. Hence, em wave is a transverse wave.

Using III Maxwell eq.ⁿ, $\boxed{\vec{\nabla} \times \vec{E} = i \vec{k} \times \vec{E}} = -\frac{\partial \vec{B}}{\partial t}$

and $\boxed{\vec{\nabla} \times \vec{H} = i \vec{k} \times \vec{H}} = \frac{\partial \vec{D}}{\partial t}$

$$\boxed{i \vec{k} \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}}$$

and

$$\boxed{i \vec{k} \times \vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t}}$$

$$H = H_0 e^{i k x - i \omega t}$$

$$\frac{\partial H}{\partial t} = -i \omega H$$

$$i \vec{k} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

$$i \vec{k} \times \vec{E} = +\mu_0 \times i \omega \vec{H}$$

$$\boxed{\vec{k} \times \vec{E} = \mu_0 \omega \vec{H}}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$; \quad \vec{E} = E_0 e^{i k x - i \omega t}$$

$$\frac{\partial E}{\partial t} = -i \omega E$$

$$i \vec{k} \times \vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \epsilon_0 \times -i \omega \vec{E}$$

$$\boxed{\vec{k} \times \vec{H} = -\epsilon_0 \omega \vec{E}}$$

Condition of orthogonality: $\vec{E}$ and $\vec{H}$ always make 90° with the dir.ⁿ of wave propagation. $\vec{E}$ and $\vec{H}$ are mutually perpendicular too.

$$\vec{H} = \frac{1}{\mu_0 \omega} (\vec{E} \times \hat{n})$$

$$= \frac{1}{\mu_0 \omega} \times k (\hat{n} \times \vec{E})$$

$$\vec{H} = \frac{1}{\mu_0 \omega} \times \frac{\omega}{c} (\hat{n} \times \vec{E})$$

$$\vec{H} = \frac{1}{\mu_0 c} (\hat{n} \times \vec{E})$$

$$|\vec{H}| = \frac{1}{\mu_0 c} |\vec{E}|$$

$$\frac{|\vec{E}|}{|\vec{H}|} = \mu_0 c$$

$$\frac{|\vec{E}|}{|\vec{H}|} = \mu_0 \cdot \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

resistance of $\rightarrow$ the medium
(impedance of medium)

$$\frac{|\vec{E}|}{|\vec{H}|} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.6 \text{ ohm}$$

impedance, $Z_0 = \frac{|\vec{E}|}{|\vec{H}|} = 376.6 \Omega$

$\rightarrow$ since $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is +ve and real so, $\vec{E}$ and $\vec{H}$ are in same phase. If Z_0 is -ve, $\vec{E}$ and $\vec{H}$ are in diff. phase.

Poynting Vector ($\vec{S}$)

$$\vec{S} = \vec{E} \times \vec{H}$$

$$= \frac{1}{\mu_0 c} \vec{E} \times (\hat{n} \times \vec{E})$$

$$\vec{S} = \frac{1}{\mu_0 c} [(\vec{E} \cdot \vec{E}) \hat{n} - (\vec{E} \cdot \hat{n}) \vec{E}]$$

$\nearrow$ x-dir.
 $\searrow$ y-dir.

$$\vec{S} = \frac{1}{\mu_0 c} E^2 \hat{n}$$

avg. of Poynting vector, $\langle \vec{S} \rangle = \frac{1}{\mu_0 c} \langle E^2 \rangle \hat{n}_{\text{real}}$

$$\langle \vec{S} \rangle = \frac{1}{\mu_0 c} \langle E_0 e^{i(kx - \omega t)} \rangle^2 \hat{n}_{\text{real}}$$

$$\langle \vec{S} \rangle = \frac{1}{\mu_0 c} E_0^2 \langle \cos^2(kx - \omega t) \rangle \hat{n}$$

$$\langle \vec{S} \rangle = \frac{1}{2\mu_0 c} E_0^2 \hat{n} \quad [\because \text{avg. value of } \cos^2 x = \frac{1}{2}]$$

$$\langle \vec{S} \rangle = \frac{1}{2Z_0} E_0^2 \hat{n}$$

$$\langle \vec{S} \rangle = \frac{1}{Z_0} E_{\text{rms}}^2 \hat{n} \rightarrow \text{average intensity}$$

Electrostatic and Magnetostatic density

$$U_e = \frac{1}{2} \epsilon_0 E^2 \rightarrow \text{electrostatic density}$$

$$U_m = \frac{1}{2} \mu_0 H^2 \rightarrow \text{magnetostatic density}$$

$$\frac{U_e}{U_m} = \frac{\epsilon_0}{\mu_0} \cdot \frac{\mu_0}{\epsilon_0} = 1$$

$$\text{So, } \boxed{U_e = U_m} \rightarrow \text{for free space}$$

$$\text{Total density of the medium} = 2U_e = 2 \times \frac{1}{2} \epsilon_0 E^2 = \epsilon_0 E^2$$

$$\langle U \rangle = \epsilon_0 \langle E^2 \rangle = \epsilon_0 \langle E_0 e^{ikx - i\omega t} \rangle$$

$$\langle U \rangle = \frac{1}{2} \epsilon_0 E^2$$

$$\frac{\langle S \rangle}{\langle U \rangle} = \frac{\frac{1}{2Z_0} E_0^2 \hat{n}}{\frac{1}{2} \epsilon_0 E_0^2} = \frac{1}{Z_0 \epsilon_0} \hat{n} = c \hat{n}$$

- So, energy in the medium is transferred in the dir.ⁿ of wave propagation at the speed of light, 3×10^8 m/s.

(ii) EM wave equation in non-conducting medium: (isotropic dielectric medium)

$$(\rho=0, \sigma=0)$$

$$\mu = \mu_r \mu_0$$

$\mu_r \rightarrow$ relative permeability

$$\epsilon = \epsilon_r \epsilon_0$$

$\epsilon_r \rightarrow$ relative permittivity

$$\nabla^2 E - \mu \epsilon \frac{\partial^2 E}{\partial t^2} = 0$$

$$\nabla^2 H - \mu \epsilon \frac{\partial^2 H}{\partial t^2} = 0$$

$$\nabla^2 \psi - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0$$

$$\begin{cases} \vec{\nabla} \cdot \vec{E} = 0 \\ \vec{\nabla} \cdot \vec{H} = 0 \\ \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{H} = 0 + \frac{\partial \vec{D}}{\partial t} \end{cases}$$

Comparing, $v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}}$

$$v = \frac{1}{\sqrt{\mu_r \epsilon_r}} \cdot \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$v = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

$\sqrt{\mu_r \epsilon_r} = n \rightarrow$ refractive index of medium

$$v = \frac{c}{n}$$

• if medium is non-magnetic, $\mu_r = 1$ so, $n = \sqrt{\epsilon_r}$

$$\nabla^2 E - \frac{1}{v^2} \frac{\partial^2 E}{\partial t^2} = 0 \quad ; \quad E(x,t) = E_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$\nabla^2 H - \frac{1}{v^2} \frac{\partial^2 H}{\partial t^2} = 0 \quad ; \quad H(x,t) = H_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$\nabla^2 \psi - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0 \quad ; \quad \boxed{\vec{k} = k \hat{n} = \frac{2\pi}{\lambda} \hat{n} = \frac{2\pi}{v} \hat{n} = \frac{\omega}{v} \hat{n}}$$

$\omega \rightarrow$ angular velocity

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} = 0 ; \vec{\nabla} \cdot \vec{E} = i\vec{k} \cdot \vec{E}$$

$$\vec{\nabla} \cdot \vec{H} = 0 ; \vec{\nabla} \cdot \vec{H} = i\vec{k} \cdot \vec{H}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} ; \vec{\nabla} \times \vec{E} = i\vec{k} \times \vec{E}$$

$$\vec{\nabla} \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} = \frac{\partial \vec{D}}{\partial t} ; \vec{\nabla} \times \vec{H} = i\vec{k} \times \vec{H}$$

$$i\vec{k} \cdot \vec{E} = i\vec{k} \cdot \vec{H} = 0$$

$$\vec{k} \cdot \vec{E} = \vec{k} \cdot \vec{H} = 0$$

$$i\vec{k} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t} ; i\vec{k} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$i\vec{k} \times \vec{E} = -\mu (-i\omega) \vec{H}$$

$$\boxed{\vec{k} \times \vec{E} = \mu \omega \vec{H}}$$

$$i\vec{k} \times \vec{H} = \epsilon (-i\omega) \vec{E}$$

$$\boxed{\vec{k} \times \vec{H} = -\epsilon \omega \vec{E}}$$

- $\vec{E}$, $\vec{H}$ and $\vec{k}$ are all at 90° to each other, hence they are ^{the} orthogonal vectors and satisfy the condition of orthogonality.

$$\vec{H} = \frac{1}{\mu \omega} (\vec{k} \times \vec{E})$$

$$\vec{H} = \frac{1}{\mu \omega} \times R (\hat{n} \times \vec{E}) = \frac{1}{\mu \omega} \times \frac{\omega}{v} (\hat{n} \times \vec{E})$$

$$\vec{H} = \frac{1}{\mu v} (\hat{n} \times \vec{E})$$

$$\left| \frac{\vec{E}}{\vec{H}} \right| = \mu v = \mu \times \frac{1}{\sqrt{\mu \epsilon}}$$

impedance of medium,

$$\boxed{Z = \left| \frac{\vec{E}}{\vec{H}} \right| = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_m \mu_0}{\epsilon_r \epsilon_0}}}$$

$$Z = \sqrt{\frac{\mu_m}{\epsilon_r}} \times \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$\boxed{Z = \sqrt{\frac{\mu_m}{\epsilon_r}} \times Z_0} , Z_0 = 376.6 \Omega$$

Z is real and +ve so that $\vec{k}$ and $\vec{H}$ are in same phase.

Poynting Vector ($\vec{S}$)

$$\vec{S} = (\vec{E} \times \vec{H})$$

$$\vec{S} = \frac{1}{\mu_0} [\vec{E} \times (\hat{n} \times \vec{E})]$$

$$\vec{S} = \frac{1}{\mu_0} [(\vec{E} \cdot \vec{E}) \hat{n} - (\vec{E} \cdot \hat{n}) \vec{E}]$$

$$\vec{S} = \frac{1}{\mu_0} \vec{E}^2 \hat{n} = \frac{1}{2} \vec{E}^2 \hat{n}$$

$$\langle \vec{S} \rangle = \frac{1}{2} \langle E^2 \rangle \hat{n}$$

$$\langle \vec{S} \rangle = \frac{1}{2} E_0^2 \langle \cos^2(\vec{k} \cdot \vec{r} - \omega t) \rangle \hat{n}$$

$$\langle \vec{S} \rangle = \frac{1}{2} E_0^2 \hat{n}$$

$$\langle \vec{S} \rangle = \frac{1}{2} E_{rms}^2 \hat{n}$$

Electrostatic and Magnetostatic density

Electrostatic density, $U_m = \frac{1}{2} \mu H^2$

$$U_e = \frac{1}{2} \epsilon E^2$$

$$\frac{U_m}{U_e} = \frac{\mu}{\epsilon} \cdot \frac{H^2}{E^2} \cdot \frac{\epsilon}{\mu} = 1$$

$$U_m = U_e$$

Total density of the medium, $u = 2U_e = 2 \times \frac{1}{2} \epsilon E^2$

$$u = \epsilon E^2$$

$$\langle \vec{u} \rangle = \epsilon \langle E^2 \rangle$$

$$\langle \vec{u} \rangle = \frac{1}{2} \epsilon E_0^2$$

$$\frac{\langle \vec{S} \rangle}{\langle \vec{u} \rangle} = \frac{\frac{1}{2} E_0^2 \hat{n}}{\frac{1}{2} \epsilon E_0^2} = \frac{1}{\epsilon} \hat{n} = v \hat{n}$$

- So energy is transferred in the medium in the dir.ⁿ of propagation of wave.

(iii) EM wave equation in conducting medium : ($B = \mu H$, $D = \epsilon E$, $J = \sigma E$, $\rho = 0$, $\sigma \neq 0$)

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \times \vec{H} = J + \delta \vec{B} / \delta t$$

$$\vec{\nabla} \cdot \vec{H} = 0$$

$$\vec{\nabla} \times \vec{E} = -\delta \vec{B} / \delta t$$

$$\nabla^2 E - \mu \sigma \frac{\delta E}{\delta t} - \mu \epsilon \frac{\delta^2 E}{\delta t^2} = 0 \quad \text{--- (1)} ; E = E_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$\nabla^2 H - \mu \sigma \frac{\delta H}{\delta t} - \mu \epsilon \frac{\delta^2 H}{\delta t^2} = 0 \quad \text{--- (2)} ; H = H_0 e^{i\vec{k} \cdot \vec{r} - i\omega t} ; \vec{k} \neq \frac{2\pi \hat{n}}{\lambda}$$

$$\nabla^2 \psi - \mu \sigma \frac{\delta \psi}{\delta t} - \mu \epsilon \frac{\delta^2 \psi}{\delta t^2} = 0 \quad \text{--- (3)} ; \psi = \psi_0 e^{i\vec{k} \cdot \vec{r} - i\omega t}$$

$$\# \quad \frac{\delta \psi}{\delta t} = -i\omega \psi \quad \frac{\delta^2 \psi}{\delta t^2} = +i\omega^2 \psi = -\omega^2 \psi$$

$$\# \quad \nabla^2 \psi \text{ means } \frac{\delta^2 \psi}{\delta x^2} = i^2 k^2 \psi = -k^2 \psi$$

Putting these values in eq. (3),

$$[-k^2 + i\omega \mu \sigma + \omega^2 \mu \epsilon] \psi = 0$$

$$\text{So, } k^2 - i\omega \mu \sigma - \omega^2 \mu \epsilon = 0$$

$$k^2 = \mu \epsilon \omega^2 + i\omega \mu \sigma$$

$$k^2 = \mu \epsilon \omega^2 \left[1 + \frac{i\sigma}{\omega \epsilon} \right]$$

$$\text{Let } k = \alpha + i\beta \Rightarrow k^2 = \alpha^2 - \beta^2 + 2i\alpha\beta$$

$$\text{Comparing, } \alpha^2 - \beta^2 = \mu \epsilon \omega^2$$

$$2\alpha\beta = \omega \mu \sigma \Rightarrow \beta = \omega \mu \sigma / 2\alpha$$

$$\text{So, } \alpha^2 - \frac{\omega^2 \mu^2 \sigma^2}{4\alpha^2} = \mu \epsilon \omega^2$$

$$4\alpha^4 - \omega^2 \mu^2 \sigma^2 = 4\mu \epsilon \omega^2 \alpha^2$$

$$4\alpha^4 - 4\mu \epsilon \omega^2 \alpha^2 - \omega^2 \mu^2 \sigma^2 = 0$$

$$\alpha^2 = \frac{4\mu\epsilon\omega^2 \pm \sqrt{16\mu^2\epsilon^2\omega^4 + 16\omega^2\mu^2\sigma^2}}{8}$$

$$\alpha^2 = \frac{4\mu\epsilon\omega^2 \pm \sqrt{16\mu^2\omega^2(\epsilon^2\omega^2 + \sigma^2)}}{8}$$

$$\alpha^2 = \frac{4\mu\epsilon\omega^2 \pm 4\mu\omega\sqrt{\epsilon^2\omega^2 + \sigma^2}}{8}$$

$$\alpha^2 = \frac{\mu\epsilon\omega^2 \pm \mu\omega\sqrt{\epsilon^2\omega^2 + \sigma^2}}{2}$$

$$\alpha^2 = \mu\epsilon\omega^2 \left[\frac{1 + \frac{1}{\omega\epsilon}(\epsilon\omega) \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}}{2} \right]$$

$$\alpha = \omega\sqrt{\mu\epsilon} \left[\frac{1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2}}{2} \right]^{1/2}$$

$$\text{So, } \beta = \omega\sqrt{\mu\epsilon} \left[\frac{\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} - 1}{2} \right]^{1/2}$$

$$\text{So, } \psi = \psi_0 e^{i(\vec{\alpha} + \vec{\beta}) \cdot \vec{r} - i\omega t}$$

$$\psi = \psi_0 e^{-\vec{\beta} \cdot \vec{r}} e^{i\vec{\alpha} \cdot \vec{r} - i\omega t}$$

$e^{-\vec{\beta} \cdot \vec{r}}$ → absorption coefficient (decreases exponentially)
(or attenuation coefficient)

ψ_0 → amplitude

α → becomes the wave no. here

$$\text{Hence } \vec{K} = \alpha \quad \vec{K} = \frac{\omega}{v} \hat{n}$$

↳ wave no.

$$\vec{v} = \frac{\omega}{K} = \frac{\omega}{\alpha}$$

Case 1: For poor conductor, $\frac{\sigma}{\omega\epsilon} \ll 1$

$$\text{So, } \alpha = \omega\sqrt{\mu\epsilon} \left[\frac{\left(1 + \frac{\sigma^2}{2\omega^2\epsilon^2}\right) + 1}{2} \right]^{1/2} = \omega\sqrt{\mu\epsilon}$$

$$\beta = \omega\sqrt{\mu\epsilon} \times \frac{\sigma}{2\omega\epsilon} = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$$

$$\text{So, } k = \alpha + i\beta = \frac{\omega\sqrt{\mu\epsilon}}{2} + \frac{i\sigma\sqrt{\mu}}{2\sqrt{\epsilon}}$$

Case 2: For good conductor, $\frac{\sigma}{\omega\epsilon} \gg 1$

$$\alpha \approx \beta = \omega\sqrt{\mu\epsilon} \left[\frac{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}{2} \right]^{1/2} = \omega\sqrt{\mu\epsilon} \times \sqrt{\frac{\sigma}{2\omega\epsilon}} = \frac{\sqrt{\omega\mu\sigma}}{\sqrt{2}}$$

$$\text{So, } k = \alpha + i\beta = (1+i) \frac{\sqrt{\omega\mu\sigma}}{2}$$

Skin depth or Penetration depth (δ)

It is the measure of penetration of an em wave in a conducting medium.

$$\delta = \frac{1}{\beta} = \frac{1}{\frac{\sqrt{\omega\mu\sigma}}{\sqrt{2}}} \rightarrow \text{for good conductors}$$

$\beta \rightarrow$ degree of attenuation

- Q. Calculate the magnitude of Poynting vector at the surface of the sun given that power radiated by sun is 3.8×10^{26} Watt and radius of sun 7×10^8 metre.

Solⁿ

$$S = \frac{P}{4\pi r^2}$$

$$S = \frac{3.8 \times 10^{26}}{4 \times \frac{22}{7} \times 7 \times 7 \times 10^{16}} = 6.7 \times 10^{10-3} = 6.7 \times 10^7 \text{ Watt/m}^2$$

- Q. If the average distance b/w the Sun and earth is 1.5×10^{11} m, show that the avg. solar energy incident on the earth is $\frac{2 \text{ cal/cm}^2\text{-min.}}{\text{solar constant}}$.

Solⁿ

$$S = \frac{3.8 \times 10^{26}}{4 \times \frac{22}{7} \times (1.5 \times 10^{11})^2} \text{ Watt/m}^2 = 0.134 \times 10^4 \text{ Watt/m}^2$$

$$1 \text{ cal} = 4.2 \text{ Joule}$$

$$\Rightarrow 0.134 \times 10^4 \text{ J/m}^2\text{-sec} = \underset{\text{convert}}{2 \text{ cal/cm}^2\text{-min}}$$

- Q. If the earth receives 2 cal/min-cm^2 solar energy, what are the magnitudes of electric and magnetic field of radiation?

Sol.ⁿ

$$S = 2 \text{ cal/min-cm}^2$$

$$S = 1400 \text{ Watt/m}^2$$

$$\vec{S} = \vec{E} \times \vec{H} = EH \sin 90^\circ \hat{n}$$

$$|\vec{S}| = EH = 1400 \text{ Watt/m}^2$$

$$\text{also, } Z_0 = \frac{E}{H} = 376.6 \text{ ohm}$$

$$\text{After solving, } E = 376.6 H$$

$$376.6 H^2 = 1400$$

$$H = 1.93 \text{ A/m}$$

$$\text{and } E = 726 \text{ V/m}$$

- Q. Assuming that all the energy from a 1000 W lamp is radiated uniformly, calculate the avg. values of the intensities of el. and mag. fields of radiation, at a distance of 2m from the lamp.

Sol.ⁿ

$$S = \frac{P}{4\pi r^2}$$

$$S = \frac{1000}{4\pi (4)} = 19.9 \text{ Watt/m}^2$$

$$EH = 19.9$$

$$\text{and, } \frac{E}{H} = 376.6$$

$$E = 376.6 H$$

$$376.6 H^2 = 19.9$$

$$H = 0.229 \text{ A/m}$$

$$\text{and } E = 86.24 \text{ V/m}$$

- Q. Find the depth of penetration of an em wave in Cu at $f = 60 \text{ Hz}$.

Sol.ⁿ

$$\delta = \sqrt{\frac{2}{\sigma \omega \mu}}$$

$$(\sigma = 5.8 \times 10^7 \text{ mho/m}; \mu_r = \mu_0 = 1)$$

$$\omega \rightarrow \text{angular vel} = 2\pi f = 120\pi$$

$$\delta = \frac{2}{\sqrt{5.8 \times 10^7 \times 120 \times 1 \times 4\pi \times 10^{-7}}} = 8.53 \text{ mm}$$

$\downarrow$ $\downarrow$
 μ_m μ_0

Q. A plane em wave travelling in the z dir.ⁿ in an unbounded lossless dielectric medium with relative permeability $\mu_r = 1$ and relative permittivity $\epsilon_r = 3$ has an el. field intensity $E = 6 \text{ V/m}$. Find

- (i) speed of em wave in the given medium
- (ii) impedance of the medium

Sol.ⁿ (i)

$$v = \frac{c}{\sqrt{\mu_r \epsilon_r}} = \frac{3 \times 10^8}{\sqrt{1 \times 3}} = \sqrt{3} \times 10^8 = 1.73 \times 10^8 \text{ m/s}$$

(ii)

$$Z = \frac{\mu_r}{\sqrt{\epsilon_r}} \times 376.6 = \frac{376.6}{\sqrt{3}} = 217 \text{ ohm}$$

$$\vec{\nabla} \cdot \vec{E} = i\vec{k} \cdot \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{H} = i\vec{k} \cdot \vec{H} = 0$$

$$\vec{\nabla} \times \vec{E} = i\vec{k} \times \vec{E} = i\mu\omega\vec{H}$$

$$\vec{\nabla} \times \vec{H} = i\vec{k} \times \vec{H} = -(i\sigma + \omega\epsilon)\vec{E}$$

$$\vec{H} = \frac{1}{\mu\omega} (\vec{k} \times \vec{E}) = \frac{1}{\mu\omega} (|\alpha + i\beta|) (\hat{n} \times \vec{E})$$

$$\left| \frac{\vec{H}}{\vec{E}} \right| = \frac{|\alpha + i\beta|}{\mu\omega} ; \vec{k} = |\alpha + i\beta| e^{i\phi} \hat{n} \quad (\text{if there is some phase difference})$$

$$|\alpha + i\beta| = \sqrt{\alpha^2 + \beta^2}$$

$$\phi = \tan^{-1} \left(\frac{\beta}{\alpha} \right)$$

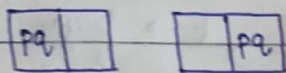
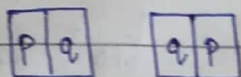
$$\sqrt{\alpha^2 + \beta^2} = \omega\sqrt{\mu\epsilon} \left[\frac{\sqrt{1 + \left(\frac{\sigma}{\mu\epsilon}\right)^2} + \sqrt{1 + \left(\frac{\sigma}{\mu\epsilon}\right)^2}}{2} \right]^{1/2}$$

$$= \omega\sqrt{\mu\epsilon} \left[1 + \left(\frac{\sigma}{\mu\epsilon}\right)^2 \right]^{1/4}$$

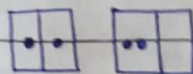
UNIT-2

Statistical Physics

Max Boltzmann



Bose Einstein



Fermi Dirac



- Molecules of gases can be explained through Max Boltzmann.

- Spin is not defined

- Identical particles Bosons (Neutrons and protons)

- Spin = 0, ± 1

- Acc. to Fermi Dirac, the particles will follow Pauli Exclusion Principle.

- Fermions - Photons

- Spin = $\pm \frac{1}{2}$

System	Max Boltzmann	Bose Einstein	Fermi Dirac
applies to the system of	identical, distinguishable particles	identical, indistinguishable particles, do not obey Pauli Exclusion Principle	identical, indistinguishable particles, obey Pauli Exclusion Principle
category of particle	classical (gas molecules)	Bosons (photons etc.)	Fermions (photons, neutrons, electrons)
examples	gas molecules	photons in cavity, liquid He at low temp.	Fermi electrons in metals, protons, neutrons
properties of particles	wave function do not overlap any spin	spin zero, ± 1 wave functions are symmetric to interchange of particles $\Psi(1,2) = \Psi(2,1)$	spin is $\pm \frac{1}{2}$ wave function antisymmetric to interchange of particles $\Psi(1,2) = -\Psi(2,1)$
distribution of particles	$n_i = \frac{g_i}{e^{\alpha + \beta E_i}}$	$n_i = \frac{g_i}{e^{\alpha + \beta E_i} - 1}$	$n_i = \frac{g_i}{e^{\alpha + \beta E_i} + 1}$

- properties of distribution of particles
 - no limit of no. of particles per state
 - no limit of no. of particles per state
 - never more than 1 particle per state

- distribution function

$$f_{MB} = \frac{1}{e^{\alpha + \beta E_i}}$$

$$f_{BE} = \frac{1}{e^{\alpha + \beta E_i} - 1}$$

$$f_{FD} = \frac{1}{e^{\alpha + \beta E_i} + 1}$$

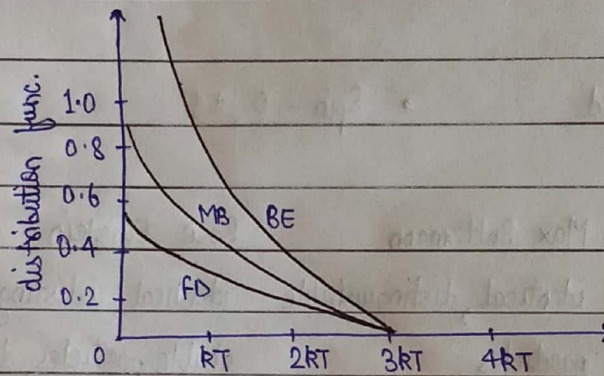
where $\alpha = -\frac{E}{RT}$, $\beta = \frac{1}{RT}$, $E \rightarrow$ fermi energy level

$E_i \rightarrow$ energy of g_i state, $g_i \rightarrow$ any energy state

$$f_{MB} = \frac{1}{e^{(E_i - E)/RT}}$$

$$f_{BE} = \frac{1}{e^{(E_i - E)/RT} - 1}$$

$$f_{FD} = \frac{1}{e^{(E_i - E)/RT} + 1}$$



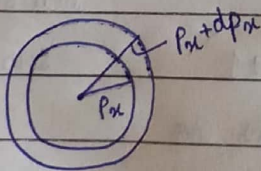
Planck's Radiation Formula

Bose - Einstein:

$$n_i = \frac{g_i}{e^{\alpha + \beta E_i} - 1}$$

$$\Delta n \cdot \Delta p \approx h$$

$$g(p) dp = \frac{\int dndydz \cdot dp_x dp_y dp_z}{h^3}$$



$$g(p) dp = \frac{v \int dp_x dp_y dp_z}{h^3}$$

$$dV = 4\pi p^2 dp$$

$$g(p) dp = \frac{4\pi v p^2 dp}{h^3}$$

$$p = \frac{h\nu}{c}$$

$$g(\nu) h\nu = \frac{4\pi v \left(\frac{h\nu}{c}\right)^2 \cdot \frac{h}{c} d\nu}{h^3}$$

$$dp = \frac{h}{c} d\nu$$

$$g(\nu) d\nu = \frac{4\pi V \nu^2}{c^3} d\nu$$

as there is simultaneous change in λ with ν , so

$$g(\nu) d\nu = \frac{8\pi V \nu^2}{c^3} d\nu$$

$$n(\nu) d\nu = \frac{g(\nu) d\nu}{e^{\alpha + \beta E_i} - 1}$$

$$n(\nu) d\nu = \frac{g(\nu) d\nu}{e^{(E_i - E)/kT} - 1}$$

Bose Einstein:

$$n_i = \frac{g_i}{e^{\alpha + \beta E_i} - 1}$$

$$n(\nu) d\nu = \frac{g(\nu) d\nu}{e^{\alpha + \beta E_i} - 1}$$

$$n(\nu) d\nu = \frac{g(\nu) d\nu}{e^{h\nu/kT} - 1}$$

$$[\because \alpha + \beta E_i = \frac{-E}{kT} + \frac{E_i}{kT} = \frac{(E_i - E)}{kT} = \frac{h\nu}{kT}$$

$$n(\nu) d\nu = \frac{8\pi V \nu^2}{c^3} \cdot \frac{d\nu}{e^{h\nu/kT} - 1}$$

$$\frac{n(\nu) d\nu}{V} = \frac{8\pi \nu^2}{c^3} \times \frac{d\nu}{e^{h\nu/kT} - 1}$$

energy per particle = $h\nu$

Total energy,

$$E(\nu) d\nu = \frac{n(\nu) d\nu}{V} \times h\nu$$

$$E(\nu) d\nu = \frac{8\pi \nu^2}{c^3} \times \frac{h\nu d\nu}{e^{h\nu/kT} - 1}$$

Planck's Radiation law,

$$E(\nu) d\nu = \frac{8\pi h \nu^3}{c^3} \times \frac{d\nu}{e^{h\nu/kT} - 1}$$

- at high temperature, $E(\nu)d\nu = \frac{8\pi h \nu^3}{c^3} \cdot \frac{d\nu}{\left[1 + \frac{h\nu}{kT}\right]}$ (binomial exp.)

$$= \frac{8\pi h \nu^3}{c^3} \times \frac{d\nu}{h\nu} \times kT$$

Rayleigh Jeans law, $E(\nu)d\nu = \frac{8\pi \nu^2 kT}{c^3} d\nu$ — (1)

- at low temperature, replace $\nu = \frac{c}{\lambda}$ and $d\nu = -\frac{c}{\lambda^2} d\lambda$ in eq. (1)

$$E\left(\frac{c}{\lambda}\right) \left(-\frac{c}{\lambda^2}\right) d\lambda = \frac{\frac{8\pi c^2}{\lambda^2} kT}{c^3} \cdot \frac{-c}{\lambda^2} d\lambda$$

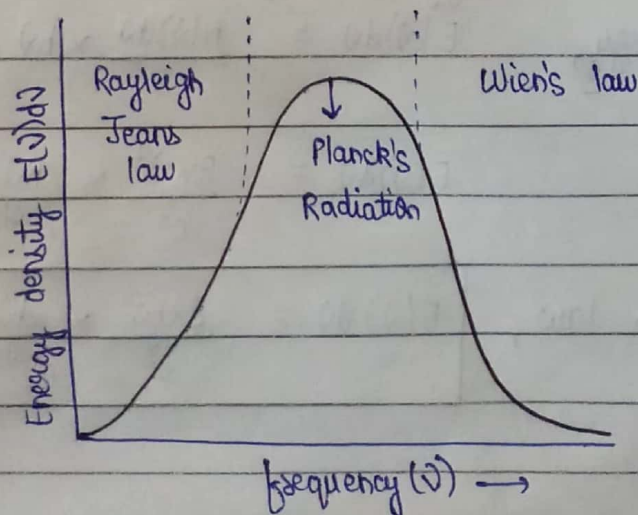
Wien's displacement law, $E(\lambda)d\lambda = \frac{-8\pi hc}{\lambda^5} \times \frac{d\lambda}{e^{hc/\lambda kT} - 1}$

- for $\lambda_{\max}$, $\frac{dE\lambda}{d\lambda} = 0$

$$\frac{hc}{\lambda_{\max} kT} = 4.965$$

$$\Rightarrow \lambda_{\max} T = \frac{hc}{4.965 k} = \text{constant}$$

Wien's displacement law, $\lambda_{\max} T = \text{constant}$



Fermi Energy

$$g(p) dp = \frac{8\pi V p^2 dp}{h^3}$$

$$\text{Energy, } E = \frac{p^2}{2m}$$

$$p = \sqrt{2mE}$$

$$dp = \frac{\sqrt{2m}}{2\sqrt{E}} dE$$

$$g(E) dE = \frac{8\pi V \cdot 2mE}{h^3} \cdot \frac{\sqrt{2m}}{2\sqrt{E}} dE$$

$$g(E) dE = \frac{8\pi V \sqrt{E} \cdot \sqrt{2} m^{3/2} dE}{h^3}$$

$$g(E) dE = \frac{8\sqrt{2} \pi V m^{3/2} \sqrt{E} dE}{h^3}$$

The no. of e^- that can have same energy

$$N = \int_0^{E_F} g(E) dE = \int_0^{E_F} \frac{8\sqrt{2} \pi V m^{3/2} \sqrt{E} dE}{h^3}$$

$$N = \frac{8\sqrt{2} \pi V m^{3/2}}{h^3} \times \frac{2}{3} E_F^{3/2}$$

$$N = \frac{16\sqrt{2} \pi V m^{3/2}}{3h^3} \times E_F^{3/2}$$

; E_F = Fermi energy

$$\frac{3Nh^3}{16\sqrt{2} \pi V m^{3/2}} = E_F^{3/2}$$

$$E_F = \frac{h^2}{2m} \left[\frac{3N}{8\pi V} \right]^{2/3}$$

Q. An e^- has a speed of 500 m/s with an accuracy of 0.005%. Calculate the uncertainty with which we can locate the position of the e^- .

Sol.ⁿ

$$\Delta x \cdot \Delta p = \frac{h}{2\pi}$$

$$500 \times \Delta x \times 9.1 \times 10^{-31} \times \frac{0.005}{100} = \frac{h}{2\pi}$$

$$\Delta x = \frac{23.09 \times 10^{-1} \times (1/500)}{100}$$

$$\Delta x = 2.309 \text{ m} \times (1/500)$$

$$\Delta x = 4.618 \text{ mm}$$

Q. Calculate the smallest possible uncertainty in the position of an e^- moving with velocity 3×10^7 m/s.

Sol.ⁿ

$$\Delta x \cdot \Delta p = \frac{h}{2\pi}$$

3×10^7 m/s is comparable to speed of light so,

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m = \frac{9.1 \times 10^{-31}}{\sqrt{1 - 0.01}} = \frac{9.1 \times 10^{-31}}{0.994}$$

$$m = 9.154 \times 10^{-31} \text{ m/kg}$$

$$m \Delta x \cdot \Delta v = 1.05 \times 10^{-34}$$

$$\Delta x = \frac{1.05 \times 10^{-34}}{9.154 \times 10^{-31} \times 3 \times 10^7}$$

$$\Delta x = 0.038 \text{ \AA}$$

Q. An e^- is confined to a box of length 10^{-8} m. Calculate the minimum uncertainty in its velocity.

Sol.ⁿ

$$m \Delta x \cdot \Delta v = 1.05 \times 10^{-34}$$

$$\Delta v = \frac{1.05 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^{-8}}$$

$$\Delta v = 0.115 \times 10^5 = 1.15 \times 10^4 \text{ m/s}$$

Q. Calculate the energy difference b/w the ground state and the first excited state for an e^- in 1-dimensional rigid box of length 10^{-10} m.

Solⁿ

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$E_1 = \frac{h^2}{8mL^2}$$

$$E_2 = \frac{4h^2}{8mL^2}$$

$$\Delta E = E_2 - E_1 = \frac{3h^2}{8mL^2} = \frac{3 \times (6.6 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times 10^{-20}}$$

$$\Delta E = 1.79 \times 10^{-17} \text{ J}$$

$$\Delta E = \frac{1.79 \times 10^{-17}}{1.6 \times 10^{-19}} = 111.8 \text{ eV}$$

Q. Can you observe the energy state for a ball of mass 10 g, moving in a box of length 10 cm.

Solⁿ

$$E_1 = \frac{h^2}{8mL^2} = \frac{(6.6 \times 10^{-34})^2}{8 \times 10^{-2} \times 10^{-2}} = \frac{5.445 \times 10^{-68}}{10^{-4}} = 5.445 \times 10^{-64} \text{ J}$$

$$E_1 = 3.4 \times 10^{-45} \text{ eV}$$

$$E_2 = 1.36 \times 10^{-44} \text{ eV}$$

$$E_3 = 3.06 \times 10^{-44} \text{ eV}$$

Since, energy difference b/w 2 states is very less, so we can't observe a discrete spectrum. We will observe a continuous spectrum.

NOTE:

$$\text{If } \psi = x, \quad \psi^* = \psi = x$$

$$\psi = x + iy, \quad \psi^* = x - iy$$

; $\psi^* \rightarrow$ complex conjugate

$$\psi = e^{ax}, \quad \psi^* = \psi = e^{ax}$$

$$\psi = e^{ix}, \quad \psi^* = e^{-ix}$$

$$\psi = e^{(x+iy)}, \quad \psi^* = e^{(x-iy)}$$

$$\psi\psi^* = \psi^*\psi = \text{always real}$$

UNIT-2

Quantum Mechanics

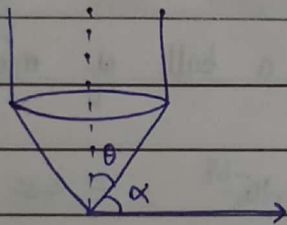
- Heisenberg's Uncertainty Principle:

$$\Delta x \cdot \Delta p = \frac{h}{2\pi} = \hbar$$

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad ; \quad \text{unit is Js}$$

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \quad \text{or} \quad \Delta J \cdot \Delta \theta \geq \frac{\hbar}{2} \quad ; \quad J \rightarrow \text{angular momentum}$$

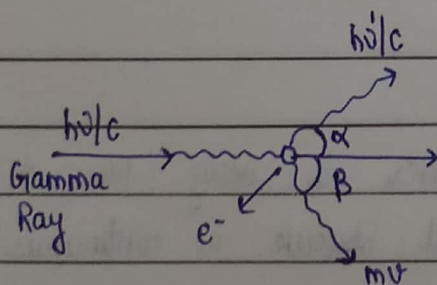
For experimental verification of eq. (1),
use Heisenberg's Gamma ray microscope,



Since we have to minimise uncertainty in position so we use gamma rays (smallest λ).

$$\Delta x = \frac{\lambda}{2 \sin \theta}$$

Resolving power



Applying conservation of momentum,

$$\frac{h\nu}{c} = \frac{h\nu'}{c} \cos \alpha + mv \cos \beta$$

$p_x \rightarrow$ momentum of e^- in x dir."

$$p_x = mv \cos \beta = \frac{h(\nu - \nu') \cos \alpha}{c}$$

$$\alpha \rightarrow (90^\circ - \theta) \text{ to } (90^\circ + \theta)$$

$$\frac{h}{c} [\nu - \nu' \cos (90^\circ - \theta)] \leq p_x \leq \frac{h}{c} [\nu - \nu' \cos (90^\circ + \theta)]$$

$$\frac{h}{c} (\nu - \nu' \sin \theta) \leq p_x \leq \frac{h}{c} (\nu + \nu' \sin \theta)$$

$$\Delta p_x = \frac{h}{c} (\nu + \nu' \sin \theta - (\nu - \nu' \sin \theta)) = \frac{2h}{c} \nu' \sin \theta$$

$$\Delta x \cdot \Delta p_x = \frac{\lambda}{2 \sin \theta} \times \frac{2h \nu \sin \theta}{c} = \frac{\lambda}{2 \sin \theta} \times \frac{2h \sin \theta}{\lambda}$$

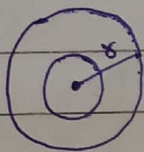
$$\Delta x \cdot \Delta p_x = \hbar$$

$$\Delta x \cdot \Delta p \geq \frac{h}{2}$$

Hence, proved.

÷ Applications of Heisenberg's Uncertainty Principle:

(i) The non-existence of e^- in nucleus:



$$r = 10^{-14} \text{ m}$$

Uncertainty is due to diameter.

$$\Delta q \cdot \Delta p \approx \hbar$$

$$\Delta q = 2 \times 10^{-14} \text{ m}$$

$$\Delta q = \Delta x$$

$$\Delta E = \frac{\Delta p^2}{2m}$$

$$\Delta p \approx \frac{6.6 \times 10^{-34}}{2 \times 3.14 \times 2 \times 10^{-14}}$$

$$\Delta p = 5.275 \times 10^{-21} \text{ kg-m/sec}$$

$$\Delta E = \frac{\Delta p^2}{2m} = \frac{(5.275 \times 10^{-21})^2}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}} = 9.7 \times 10^7 \text{ eV}$$

$$\Delta E = 9.7 \times 10^7 \text{ eV}$$

$$\Delta E = 97 \text{ MeV}$$

But experimentally, max. energy of an e^- is 4 MeV.

$$T_{\text{exp}} = 4 \text{ MeV}$$

(ii) Energy of particle in 1-D box:

$$\Delta x \cdot \Delta p = \hbar$$

$$\Delta p = \frac{\hbar}{\Delta x} = \frac{\hbar}{l}$$

; $l \rightarrow$ length of box

$$K.E. = \frac{\hbar^2}{2ml^2}$$

$$\therefore K.E. = \frac{p^2}{2m}$$

(iii) Minimum energy of a Harmonic oscillator:

$$E = K.E. + P.E.$$

$$= \frac{p^2}{2m} + \frac{1}{2} K (\Delta x)^2$$

$$; \Delta x \cdot \Delta p \approx \frac{\hbar}{2}$$

$$E = \frac{\hbar^2}{8m(\Delta x)^2} + \frac{1}{2} K (\Delta x)^2$$

for minima or maxima, $\frac{dE}{d(\Delta x)} = 0$

$$\frac{dE}{d(\Delta x)} = \frac{\hbar^2 (-2)}{8m(\Delta x)^3} + K \Delta x = 0$$

$$\frac{\hbar^2}{4m(\Delta x)^3} = K \Delta x$$

$$(\Delta x)^4 = \frac{\hbar^2}{4mK}$$

$$\Delta x = \left(\frac{\hbar^2}{4mK} \right)^{1/4} \rightarrow \text{minima}$$

$$E = \frac{\hbar^2}{8m \left(\frac{\hbar^2}{4mK} \right)^{1/2}} + \frac{1}{2} K \left(\frac{\hbar^2}{4mK} \right)^{1/2}$$

$$E = \frac{\hbar \sqrt{K}}{4\sqrt{m}} + \frac{\sqrt{K} \hbar}{2 \times 2\sqrt{m}} \Rightarrow E = \frac{\hbar \sqrt{K}}{2\sqrt{m}}$$

$$E = \frac{1}{2} \hbar \omega \quad ; \quad \omega = \sqrt{\frac{K}{m}} \rightarrow \text{angular frequency of harmonic oscillator}$$

$$E = \frac{1}{2} \hbar 2\pi \nu$$

$$E = \frac{\hbar \nu}{2}$$

• De-Broglie Hypothesis:

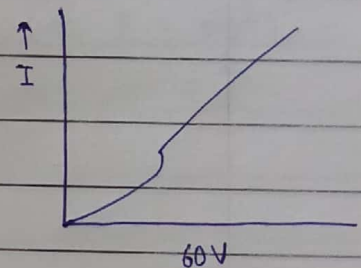
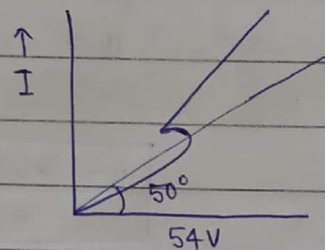
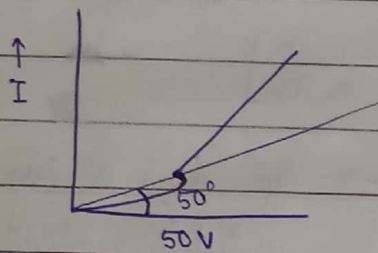
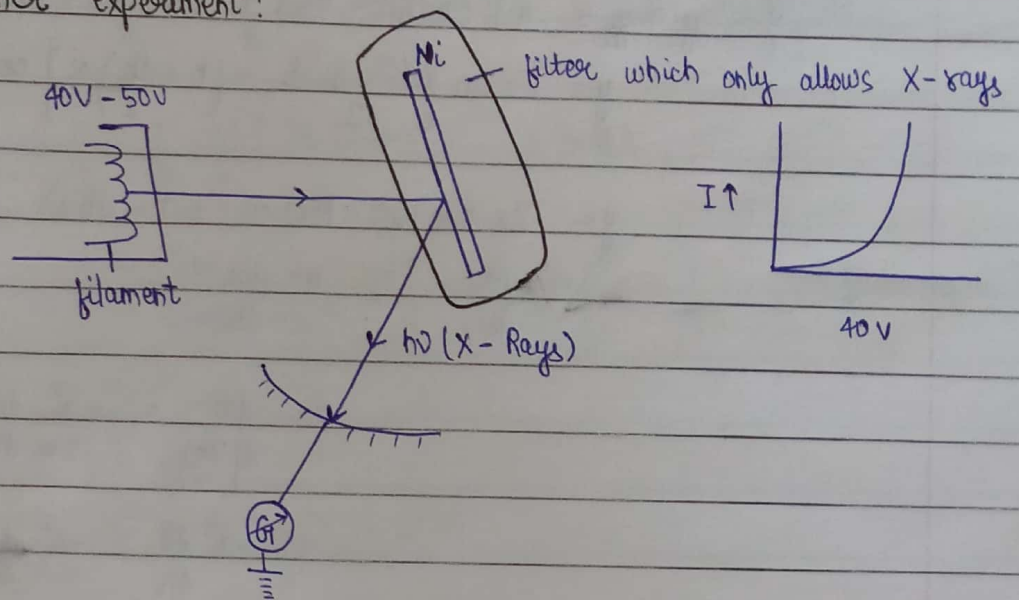
$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE_k}}$$

$$E_k = qV$$

$$\text{So, } \lambda = \frac{h}{\sqrt{2mqV}} = \frac{12.3 \text{ \AA}}{\sqrt{V}}$$

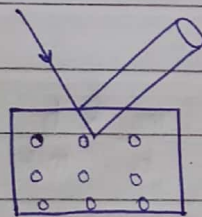
$$\text{or } \lambda = \frac{h}{\sqrt{2mkT}} \quad ; \quad K \rightarrow \text{Boltzmann constant} \\ T \rightarrow \text{temperature}$$

Davisson and Germer experiment:



$$\lambda = \frac{12.3}{\sqrt{54}} = 1.66 \text{ \AA}$$

Bragg's Law:



$$2d \sin \theta = n\lambda$$

Structure - FCC

Miller indices - (1,1,1)

$d = 0.91 \text{ \AA}$ (for Nickel)

for $n=1$, $\theta = 65^\circ$, $\lambda = 1.65 \text{ \AA}$

Since $\lambda = 1.65 \text{ \AA}$ (approx.) in both cases, it proves particles follow wave nature and are in the form of wave packets called photons.

Phase velocity and Group velocity:

(v)

(u)

Velocity of single wave is called phase velocity.

When waves are propagating together, their resultant of their phase velocities on superimposition is called group velocity.

$$y_1 = a \sin(\omega t - kx)$$

$$y_2 = a \sin[(\omega + d\omega)t - (k + dk)x]$$

$$y = y_1 + y_2 = a [\sin(\omega t - kx) + \sin\{(\omega + d\omega)t - (k + dk)x\}]$$

$$y = 2a \sin \left[\frac{(2\omega + d\omega)t - (k + dk)x}{2} \right] \cos(\omega t - kx)$$

$$y = 2a \sin(\omega t - kx) \cos(\omega t - kx)$$

$$\frac{\omega + d\omega}{2} \approx \omega$$

$$v = \frac{\omega}{k}$$

$$u = \frac{d\omega}{dk}$$

$$k + \frac{dk}{2} \approx k$$

$$u = \frac{d\omega}{d\left(\frac{2\pi}{\lambda}\right)} = \frac{-\lambda^2}{2\pi} \frac{d\omega}{d\lambda} = \frac{-\lambda^2}{2\pi} \frac{d(2\pi\nu)}{d\lambda}$$

$$u = \frac{-\lambda^2}{d\lambda} \frac{d\nu}{d\lambda} = \frac{-\lambda^2}{d\lambda} \frac{d\left(\frac{v}{\lambda}\right)}{d\lambda}$$

$$u = -\lambda^2 \left[\frac{-v}{\lambda^2} + \frac{1}{\lambda} \frac{dv}{d\lambda} \right]$$

$$\boxed{u = v - \lambda \frac{dv}{d\lambda}}$$

→ gp. vel. is less than individual particle's velocity.

This means particle would move ahead of wave, which is not possible.

$$\lambda = \frac{h}{mv}$$

$$E = h\nu \Rightarrow \nu = \frac{E}{h} = \frac{mc^2}{h}$$

$$u = \lambda \nu = \frac{mc^2}{h} \times \frac{h}{mv} \Rightarrow \boxed{u = \frac{c^2}{v}}$$

$\boxed{u > c}$ → which is not possible.

$$\Rightarrow u = \frac{d\omega}{dk} = \frac{d(2\pi\nu)}{d\left(\frac{2\pi}{\lambda}\right)} = \frac{d(2\pi\nu)}{d\left(\frac{2\pi\nu}{v}\right)} = \frac{d\nu}{d\left(\frac{\nu}{v}\right)}$$

$$\frac{1}{u} = \frac{d\left(\frac{\nu}{v}\right)}{d\nu} = \frac{d\left(\frac{1}{\lambda}\right)}{d\lambda}$$

$$\# \lambda = \frac{h}{mv} \Rightarrow \frac{1}{\lambda} = \frac{mv}{h}$$

$$\frac{1}{2} mv^2 = E - v$$

$$mv = [2m(E - v)]^{1/2}$$

$$v = \left(\frac{2(E - v)}{m} \right)^{1/2} \quad - (1)$$

$$\text{So, } \frac{1}{u} = \frac{1}{h} \frac{d[2m(E-V)]^{1/2}}{dV} = \frac{1}{h} \frac{d[2m(h\nu - V)]^{1/2}}{dV}$$

$$\frac{1}{u} = \frac{1}{h} \times \frac{1}{2} [2m(h\nu - V)]^{-1/2} \times 2m$$

$$\frac{1}{u} = \frac{m}{[2m(E-V)]^{1/2}} = \left[\frac{m}{2(E-V)} \right]^{1/2} = \frac{1}{v} \quad (\text{from eq. (1)})$$

Non-Relativistic ($E \neq mc^2$):

$$\lambda = \frac{h}{mv}$$

$$E = h\nu \Rightarrow \nu = \left(\frac{2h}{mv^2} \right)^{-1} = \frac{mv^2}{2h} \quad (\text{as } E = \frac{mv^2}{2})$$

$$u = \nu\lambda = \frac{mv^2}{2h} \times \frac{h}{mv}$$

$$\boxed{u = \frac{v}{2}}$$

• Schrodinger wave equation:

(i) Time independent:

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \text{--- (1)}$$

$$\psi = A \sin(\omega t - kx)$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

Solution of eq. (1) is $\psi(x, t) = \psi_0 e^{-i\omega t}$
 $\Rightarrow \frac{\partial \psi}{\partial t} = -i\omega \psi_0 e^{-i\omega t}$

$$\frac{\partial^2 \psi}{\partial t^2} = i^2 \omega^2 \psi_0 e^{-i\omega t} = -\omega^2 \psi$$

$$\therefore \nabla^2 \psi = \frac{-\omega^2 \psi}{v^2}$$

$$\nabla^2 \psi = \frac{-4\pi^2}{\lambda^2} \psi$$

$$[\because k = \frac{\omega}{v} = \frac{2\pi}{\lambda}]$$

$$\nabla^2 \psi = \frac{-4\pi^2 m^2 v^2}{h^2} \psi$$

$$[\because \lambda = \frac{h}{mv}]$$

$$\nabla^2 \psi = -\frac{m^2 v^2}{\hbar^2} \psi$$

$$[\because \hbar = \frac{h}{2\pi}]$$

$$\frac{1}{2} m v^2 = E - V$$

↓ ↗ Potential energy
Total energy

$$m^2 v^2 = 2m(E - V)$$

$$\text{So, } \nabla^2 \psi = -\frac{2m(E - V)}{\hbar^2} \psi$$

$$\boxed{\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0} \quad - (2)$$

(ii) For free particle:

$$V = 0$$

$$\text{So, } \nabla^2 \psi + \frac{2m}{\hbar^2} E \psi = 0$$

(iii) Time dependent:

$$\text{from eq. (2), } \frac{\hbar^2}{2m} \nabla^2 \psi + E \psi - V \psi = 0$$

$$\left(\frac{\hbar^2}{2m} \nabla^2 - V \right) \psi + E \psi = 0$$

$$\left(V - \frac{\hbar^2}{2m} \nabla^2 \right) \psi = E \psi \quad - (3)$$

$$\# \quad \psi = \psi_0 e^{-i\omega t}$$

$$\frac{\partial \psi}{\partial t} = -i\omega \psi = -i(2\pi\nu) \psi$$

$$E = h\nu \Rightarrow \nu = \frac{E}{h}$$

$$\frac{\partial \psi}{\partial t} = -i \frac{2\pi}{h} E \psi = -i \frac{E \psi}{\hbar} \times \frac{i}{i}$$

$$\frac{\partial \psi}{\partial t} = \frac{E \psi}{i\hbar}$$

$$E\psi = i\hbar \frac{\partial \psi}{\partial t}$$

So, from eq. (3),

$$\boxed{\left(V - \frac{\hbar^2 \nabla^2}{2m} \right) \psi = i\hbar \frac{\partial \psi}{\partial t}}$$

$\underbrace{\quad}_{H} \qquad \underbrace{\quad}_{E = i\hbar \frac{\partial}{\partial t}}$

$$H = -\frac{\hbar^2 \nabla^2}{2m} + V$$

So, $\boxed{H = E}$

$H \rightarrow$ Hamiltonian energy

OPERATORS

(i) Energy operator:

$$E_{op} = i\hbar \frac{\partial}{\partial t} \rightarrow \text{for time coordinates}$$

$$H_{op} = E_{op} = -\frac{\hbar^2 \nabla^2}{2m} + V$$

(ii) Momentum operator:

$$K.E. = \frac{p^2}{2m} = -\frac{\hbar^2 \nabla^2}{2m}$$

$$K.E. = \frac{i^2 \hbar^2 \nabla^2}{2m} = \frac{p^2}{2m}$$

So, we get $p_{op} = -i\hbar \nabla$

$$p_{x op} = -i\hbar \frac{\partial}{\partial x} ; p_{y op} = -i\hbar \frac{\partial}{\partial y} ; p_{z op} = -i\hbar \frac{\partial}{\partial z}$$

(iii) Kinetic energy operator:

$$T_{op} = -\frac{\hbar^2 \nabla^2}{2m}$$

(iv) Velocity operator:

$$\frac{1}{2} m v^2 = -\frac{\hbar^2 \nabla^2}{2m}$$

$$v^2 = \frac{-\hbar^2 \nabla^2}{m^2} = \frac{i^2 \hbar^2 \nabla^2}{m}$$

$$\Rightarrow v_{op} = \frac{i\hbar \nabla}{m}$$

$$v_{xop} = \frac{i\hbar}{m} \frac{\partial}{\partial x} ; v_{yop} = \frac{i\hbar}{m} \frac{\partial}{\partial y} ; v_{zop} = \frac{i\hbar}{m} \frac{\partial}{\partial z}$$

(iv) Displacement operators:

$$x_{op} = x$$

$$x_{xop} = x$$

$$x_{yop} = y$$

$$x_{zop} = z$$

Meaning of Ψ

$$\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0$$

$\Psi \rightarrow$ amplitude

Intensity $\propto \Psi^2$; Probability density $\propto \Psi^2$

$$\hookrightarrow p \propto |\Psi|^2 = \Psi \Psi^*$$

$$p = \int_{-\infty}^{\infty} \Psi \Psi^* (dx) \rightarrow \text{1-dimensional space coordinate}$$

Since space is not defined, we have put the limits from $-\infty$ to ∞ .

If $\Psi = e^x$, then $\Psi^* = e^x$

$\Psi = e^{x+iy}$, then $\Psi^* = e^{x-iy}$

If $p = 1$, $\Psi \rightarrow$ Normalised wave function

Q. $\Psi = A \sin\left(\frac{2n\pi x}{L}\right)$ for $0 \leq x \leq L$, find the value of A by using

normalisation condition.

Solⁿ

$$p = \int_{-\infty}^{\infty} \Psi \Psi^* dx$$

$$1 = \int_{-\infty}^{\infty} A \sin\left(\frac{2n\pi x}{L}\right) A \sin\left(\frac{2n\pi x}{L}\right) dx = A^2 \int_{-\infty}^{\infty} \sin^2\left(\frac{2n\pi x}{L}\right) dx$$

$$1 = A^2 \int_{-\infty}^{\infty} \frac{1 - \cos\left(\frac{4n\pi x}{L}\right)}{2} dx$$

$$1 = \frac{A^2}{2} \int_0^L 1 - \cos\left(\frac{4n\pi x}{L}\right) dx$$

Since $0 \leq x \leq l$, $1 = \frac{A^2}{2} \int_0^l 1 - \cos\left(\frac{4\pi x}{l} x\right) dx$

$$1 = \frac{A^2}{2} \left[x - \frac{\sin \frac{4\pi x}{l} x}{\frac{4\pi x}{l}} \right]_0^l$$

$$1 = \frac{A^2}{2} \left[l - \frac{\sin 4\pi x \cdot l}{4\pi x} \right]$$

$$1 = \frac{A^2 l}{2} \Rightarrow A = \sqrt{\frac{2}{l}}$$

So, normalised wave function is $\psi(x) = \sqrt{\frac{2}{l}} \sin\left(\frac{2\pi x}{l} x\right)$.

Q. Normalise the following function: $\psi(x) = A e^{-\alpha x}$ for $x > 0$ and $= A e^{\alpha x}$ for $x < 0$.

Sol.

$$P = \int_{-\infty}^{\infty} \psi \psi^* dx$$

$$\begin{cases} 1 = \int_0^{\infty} A e^{-\alpha x} A e^{-\alpha x} dx \\ 1 = A^2 \int_0^{\infty} e^{-2\alpha x} dx = A^2 \left[\frac{e^{-2\alpha x}}{-2\alpha} \right]_0^{\infty} \\ 1 = \frac{A^2}{-2\alpha} [0 - 1] = \frac{A^2}{2\alpha} \\ A = \sqrt{2\alpha} \end{cases}$$

$$1 = \int_{-\infty}^0 A^2 e^{2\alpha x} dx + \int_0^{\infty} A^2 e^{-2\alpha x} dx$$

$$1 = \frac{A^2}{2\alpha} [e^{2\alpha x}]_{-\infty}^0 + \frac{A^2}{(-2\alpha)} [e^{-2\alpha x}]_0^{\infty}$$

$$1 = \frac{A^2}{2\alpha} [1 - 0] - \frac{A^2}{2\alpha} [0 - 1] = \frac{A^2}{\alpha} \Rightarrow A = \sqrt{\alpha}$$

Q. $\psi(x) = \sqrt{\frac{2}{l}} \frac{\sin \pi x}{l}$ for $0 < x < l$; $\psi(x) = 0$ elsewhere, find the

probability in the region $0 < x < \frac{l}{2}$.

Sol.ⁿ

$$P = \int_0^{l/2} \Psi \Psi^* dx$$

$$P = \int_0^{l/2} \left(\sqrt{\frac{2}{l}} \right)^2 \left(\frac{\sin \pi x}{l} \right)^2 dx$$

$$P = \frac{2}{l} \int_0^{l/2} \frac{1 - \cos 2\pi x}{2} dx$$

$$P = \frac{2}{l} \cdot \frac{1}{2} \int_0^{l/2} (1 - \cos 2\pi x) dx$$

$$P = \frac{1}{l} \left[x - \frac{\sin 2\pi x}{\frac{2\pi}{l}} \right]_0^{l/2}$$

$$P = \frac{1}{l} \left[\frac{l}{2} - \frac{\sin 2\pi l}{\frac{2\pi}{l}} \right] = \frac{1}{2}$$

Q. $\Psi(x) = N e^{-x^2/2a^2 + ikrx}$ for free particle. Find

(i) N

(ii) In what region of space, the particle is most likely to be found

Sol.ⁿ (i)

$$P = \int_{-\infty}^{\infty} \Psi \Psi^* dx$$

$$P = \int_{-\infty}^{\infty} N e^{-x^2/2a^2 + ikrx} \cdot N e^{-x^2/2a^2 - ikrx} dx$$

$$P = 2N^2 \int_0^{\infty} e^{-x^2/a^2} dx$$

$$u = x^2$$

$$du = 2x dx \Rightarrow dx = \frac{du}{2\sqrt{u}}$$

$$P = 2N^2 \int_0^{\infty} e^{-u/a^2} \cdot \frac{du}{2\sqrt{u}}$$

Since $\int_{-\infty}^{\infty} e^{-x^2/a^2} dx = a\sqrt{\pi}$

So, $1 = N^2 a\sqrt{\pi}$

$N = \frac{1}{\sqrt{a(\pi)^{1/4}}}$

(iii) $P(x) = |\psi(x)|^2 = N^2 e^{-x^2/a^2} = \frac{1}{a\sqrt{\pi}} e^{-x^2/a^2}$

So $P(x)$ is max. for $x=0$ and $\downarrow$ es exponentially for $|x| > a$.

So, desired region is $-a < x < a$.

Conditions satisfied by wave function

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

- It must be finite everywhere. If ψ is infinite for a particular point, it would mean an infinitely large probability of finding the particle at that point. It would violate the uncertainty principle.
- It must have a single value. If ψ has more than 1 value at any point, it would mean more than one value of probability of finding the particle, which is incorrect.
- It must be continuous and have a continuous first derivative everywhere. $\frac{\partial^2 \psi}{\partial x^2}$ must be finite everywhere. It can be so only if $\frac{\partial \psi}{\partial x}$ has no discontinuity.

Postulates of Quantum mechanics

- There is a state vector (wave function) ψ associated with every physical state of the system.
- For every physical observable, there is corresponding linear operator. The operators associated with various dynamical variables like energy,

momentum.

- States that only possible values which a measurement of the observable whose operator is $\hat{P}$ will give Eigen values P_n of the equation $\hat{P}\psi_n = P_n\psi_n$
- Gives the rule for extracting information from the wave function. The expectation value of a variable of a system in the state ψ is $\langle f \rangle$, which is also called average value.

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} \psi f_{op} \psi^* d\tau}{\int_{-\infty}^{\infty} \psi \psi^* d\tau}$$

Eg.

1) $\psi = x^2$

$$\hat{p} = \frac{d}{dx}$$

$$\hat{p}\psi = \frac{d(x^2)}{dx} = 2x$$

2) $\psi = e^{x^2}$

$$\hat{p} = \frac{d}{dx}$$

$$\hat{p}\psi = 2x e^{x^2}$$

$$\hat{p}\psi = P_n\psi_n$$

eigen value

eigen function

- Eigen value should be free from any variable.

#

$$P = \int_{-\infty}^{\infty} \psi \psi^* d\tau = N$$

To make it normalised, $P = \int_{-\infty}^{\infty} \frac{\psi}{\sqrt{N}} \times \frac{\psi^*}{\sqrt{N}} d\tau = 1$

Q. Calculate the expectation value of P and P^2 for the wave function

$$\psi(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L} \quad \text{for } 0 < x < L$$

$$\psi(x) = 0 \quad \text{for } |x| > L$$

Solⁿ

$$\psi(x) = \sqrt{\frac{2}{l}} \frac{\sin \pi x}{l}$$

$$p_{op} = i\hbar \frac{\partial}{\partial x}$$

$$p_{op} (\psi^*(x)) = i\hbar \frac{\partial}{\partial x} \left[\sqrt{\frac{2}{l}} \frac{\sin \pi x}{l} \right] = \frac{\pi}{l} \sqrt{\frac{2}{l}} i\hbar \cos \pi x$$

$$\langle p_x \rangle = \int_{-\infty}^{\infty} \psi p_{op} \psi^* dx$$

$$\text{But } \langle p_x \rangle = \int_0^l \sqrt{\frac{2}{l}} \frac{\sin \pi x}{l} \cdot \frac{\pi}{l} \sqrt{\frac{2}{l}} i\hbar \cos \pi x dx$$

$$\langle p_x \rangle = \frac{i\hbar}{\pi l^2} \int_0^l \sin 2\pi x dx = \frac{i\hbar}{\pi l^2} \left[-\frac{\cos 2\pi x}{2\pi} \right]_0^l \times \frac{l}{2\pi}$$

$$\langle p_x \rangle = \frac{-i\hbar}{\pi^2 l} [\cos 2\pi - \cos 0] = 0$$

$$p_{op}^2 = -\hbar^2 \frac{\partial^2}{\partial x^2}$$

$$p_{op}^2 (\psi^*(x)) = -\hbar^2 \frac{\partial^2}{\partial x^2} \left[\sqrt{\frac{2}{l}} \frac{\sin \pi x}{l} \right]$$

$$= -\hbar^2 \sqrt{\frac{2}{l}} \frac{\partial}{\partial x} \left(\frac{\pi \cos \pi x}{l} \right)$$

$$= \frac{+\hbar^2 \pi}{l} \sqrt{\frac{2}{l}} \cdot \frac{\pi \sin \pi x}{l} = \frac{\pi^2 \hbar^2}{l^2} \sqrt{\frac{2}{l}} \cdot \frac{\sin \pi x}{l}$$

$$\langle p_x^2 \rangle = \int_0^l \psi p_{op}^2 \psi^* dx$$

$$= \int_0^l \sqrt{\frac{2}{l}} \frac{\sin \pi x}{l} \cdot \frac{\pi^2 \hbar^2}{l^2} \sqrt{\frac{2}{l}} \frac{\sin \pi x}{l} dx$$

$$= \frac{2\pi^2 \hbar^2}{l^3} \int_0^l \frac{\sin^2 \pi x}{l} dx = \frac{2\pi^2 \hbar^2}{l^3} \int_0^l \frac{1 - \cos 2\pi x}{2} dx$$

$$= \frac{\pi^2 \hbar^2}{l^3} \left[x - \frac{l \sin 2\pi x}{2\pi} \right]_0^l$$

$$= \frac{\pi^2 \hbar^2}{l^3} [l - 0] = \frac{\pi^2 \hbar^2}{l^2}$$

Particle in a Box

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$V=0$, where particle exists

$$\nabla^2 \psi + \frac{2m E \psi}{\hbar^2} = 0$$

The above eq.ⁿ can be solved by a method of separation of variables.

$$\text{So, } \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m E \psi}{\hbar^2} = 0$$

$$\psi(x, y, z) = X(x) \cdot Y(y) \cdot Z(z)$$

$$\frac{\partial^2 \psi}{\partial x^2} = YZ \frac{\partial^2 X}{\partial x^2} ; \quad \frac{\partial^2 \psi}{\partial y^2} = XZ \frac{\partial^2 Y}{\partial y^2} ; \quad \frac{\partial^2 \psi}{\partial z^2} = XY \frac{\partial^2 Z}{\partial z^2}$$

$$YZ \frac{\partial^2 X}{\partial x^2} + XZ \frac{\partial^2 Y}{\partial y^2} + XY \frac{\partial^2 Z}{\partial z^2} + \frac{2m E XYZ}{\hbar^2} = 0$$

$$\text{Dividing by } XYZ, \quad \frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} + \frac{2m E}{\hbar^2} = 0$$

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} - \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} - \frac{2m E}{\hbar^2}$$

$$\text{Let } \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = \boxed{k_x} \rightarrow \text{constant term}$$

$$k_x = -\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} - \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} - \frac{2m E}{\hbar^2}$$

$$\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = -k_x - \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} - \frac{2m E}{\hbar^2} = k_y \text{ (let)}$$

$$\frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = -k_x - k_y - \frac{2m E}{\hbar^2} = k_z \text{ (let)}$$

$$\text{So, } k_x + k_y + k_z = -\frac{2m E}{\hbar^2} \quad \text{and} \quad E = E_x + E_y + E_z$$

$$k_x = -\frac{2m E_x}{\hbar^2} ; \quad k_y = -\frac{2m E_y}{\hbar^2} ; \quad k_z = -\frac{2m E_z}{\hbar^2}$$

$$\text{So, } \frac{1}{x} \frac{\partial^2 X}{\partial x^2} = -\frac{2m E_x}{\hbar^2} ; \text{ So, } \frac{1}{x} \frac{\partial^2 X}{\partial x^2} + \frac{2m E_x}{\hbar^2} = 0$$

$$\frac{1}{y} \frac{\partial^2 Y}{\partial y^2} + \frac{2m E_y}{\hbar^2} = 0 ; \quad \frac{1}{z} \frac{\partial^2 Z}{\partial z^2} + \frac{2m E_z}{\hbar^2} = 0$$

$$X(x) = A \sin(Bx + C) ; \quad 0 < x < l_x$$

$$|X(x)|^2 = 0 \quad \text{at} \quad x=0$$

$$\text{at} \quad x=l_x$$

$$\text{ii) at } x=0 ; X(x)=0 \Rightarrow 0 = A \sin(0+C) \Rightarrow \sin C=0 \Rightarrow C=0$$

$$\text{iii) at } x=l_x ; X(x)=0 \Rightarrow X(x) = A \sin Bx \Rightarrow 0 = A \sin B l_x$$

$$\Rightarrow B l_x = n_x \pi \Rightarrow B = \frac{n_x \pi}{l_x}$$

$$\text{So, we get } X(x) = A \sin\left(\frac{n_x \pi}{l_x} \cdot x\right)$$

$$\text{Probability, } P = \int_0^{l_x} |X(x)|^2 dx = 1 \Rightarrow A^2 \int_0^{l_x} \sin^2\left(\frac{n_x \pi}{l_x} \cdot x\right) dx = 1$$

$$\text{So, } A^2 \times \frac{l_x}{2} = 1 \Rightarrow A = \sqrt{\frac{2}{l_x}}$$

$$\text{So, } X(x) = \sqrt{\frac{2}{l_x}} \sin\left(\frac{n_x \pi}{l_x} \cdot x\right) \quad \text{--- (1)}$$

$$Y(y) = \sqrt{\frac{2}{l_y}} \sin\left(\frac{n_y \pi}{l_y} \cdot y\right)$$

$$Z(z) = \sqrt{\frac{2}{l_z}} \sin\left(\frac{n_z \pi}{l_z} \cdot z\right)$$

$$\Psi = XYZ$$

$$\Psi = \sqrt{\frac{2}{l_x l_y l_z}} \sin\left(\frac{n_x \pi}{l_x} \cdot x\right) \sin\left(\frac{n_y \pi}{l_y} \cdot y\right) \sin\left(\frac{n_z \pi}{l_z} \cdot z\right)$$

$$\frac{1}{x} \frac{\partial^2 X}{\partial x^2} + \frac{2m E_x}{\hbar^2} = 0 \Rightarrow \frac{\partial^2 X}{\partial x^2} + \frac{2m E_x}{\hbar^2} X = 0 \quad \text{--- (2)}$$

$$\text{from eq. (1), } \frac{\partial^2 X}{\partial x^2} = -\left(\frac{n_x \pi}{l_x}\right)^2 X(x) \rightarrow \text{putting in eq. (2)}$$

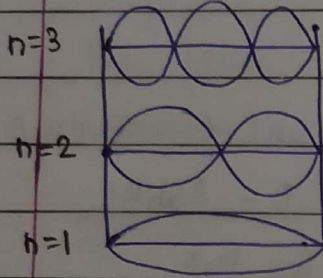
$$\left[-\left(\frac{n_x \pi}{l_x}\right)^2 + \frac{2m E_x}{\hbar^2} \right] X(x) = 0 \Rightarrow \frac{2m E_x}{\hbar^2} = \left(\frac{n_x \pi}{l_x}\right)^2$$

$$E_n = \frac{n_x^2 \pi^2}{L_x^2} \times \frac{b^2}{4\pi^2 \times 2m}$$

$$\Rightarrow E_x = \frac{n_x^2 h^2}{8m L_x^2} ; E_y = \frac{n_y^2 h^2}{8m L_y^2} ; E_z = \frac{n_z^2 h^2}{8m L_z^2}$$

and $E = E_x + E_y + E_z$

$$E = \frac{h^2}{8m} \left[\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right]$$



$n_x \rightarrow$ no. of harmonics in x-dir.ⁿ

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UNIT-3

Crystal Structure

Crystal structure: The arrangement of atoms in a crystal is in periodically repeating pattern, which represents the crystal structure.

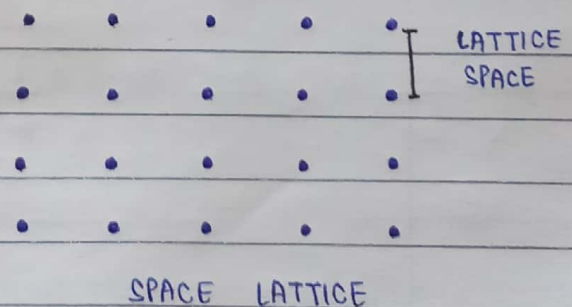
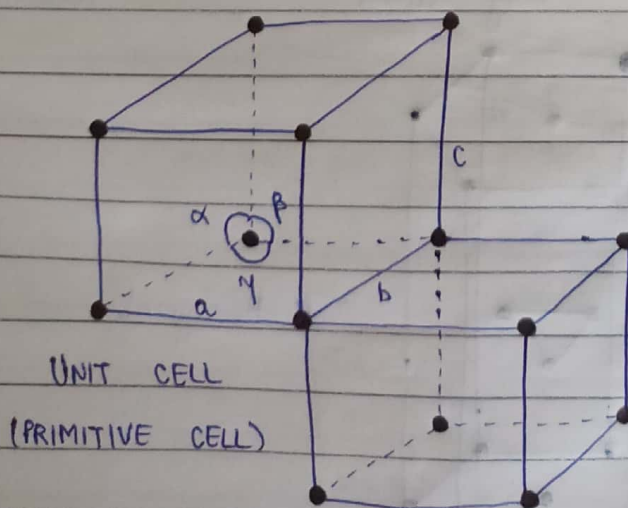
Unit cell: The metallic crystals can be considered as consisting of tiny blocks which are repeated in 3-D pattern. The tiny block formed by the arrangement of atoms is called unit cell.

Space lattice: A space lattice is defined as an array of points in 3 dimensions in which every point has surroundings identical to that of every other point in the array.

Lattice space: Space between the atoms (points) which is called interatomic spacing or lattice spacing.

Primitive cell: A primitive cell is one which has got points or atoms only at the corners of the unit cell.

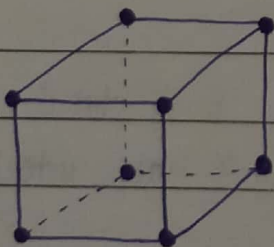
Lattice parameters of a unit cell: They are 6 in total. 3 are primitive, represented by a, b, c and other 3 are interfacial angles, represented by α, β, γ .



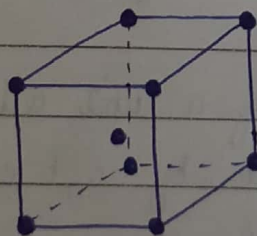
	SYSTEM	PRIMITIVES	ANGLES	EXAMPLES
1.	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	CaF_2 , NaCl
2.	Monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90^\circ \neq \gamma$	FeSO_4 , Na_2SO_4
3.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	CuSO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$
4.	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	BaSO_4 , MgSO_4
5.	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	SnO_2 , NiSO_4
6.	Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	CaSO_4
7.	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ$; $\gamma = 120^\circ$	SiO_2 , AgI

Bravais crystal lattice:

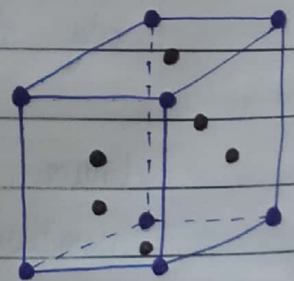
1. Cubic - SC, BCC, FCC
2. Monoclinic - Simple, Body centred
3. Triclinic - Simple
4. Tetragonal - Simple, Body centred
5. Orthorhombic - Simple, Base centred, Body centred, Face centred
6. Rhombohedral - Simple
7. Hexagonal - Simple



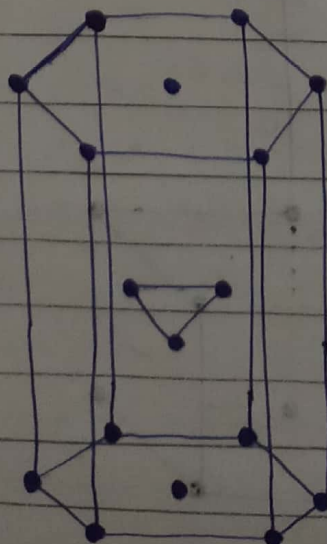
Simple



Body - centred



Face - centred



Hexagonal

Q. calculate the expectation value of p and p^2 for the wave function

$$\psi = \frac{\sqrt{2}}{l} \sin \frac{\pi x}{l} \text{ for } 0 < x < l$$

= 0 for $|x| > l$ already done

Q. $\psi = N e^{-\frac{x^2}{2a^2} + ikx}$ for free particle. calculate the expectation value of x and p .

Soln

$$p = -i\hbar \frac{d}{dx}$$

$$\psi(x) = N e^{-\frac{x^2}{2a^2} + ikx}$$

$$p_{op} |\psi^*(x)| = -i\hbar \frac{d}{dx} |N e^{-\frac{x^2}{2a^2} - ikx}|$$

$$= i\hbar N e^{-\frac{x^2}{2a^2} - ikx} \left[\frac{-x}{a^2} - ik \right]$$

$$\langle p_x \rangle = \int_{-\infty}^{\infty} \psi p_{op} \psi^* dx = \int_{-\infty}^{\infty} N e^{-\frac{x^2}{2a^2} + ikx} i\hbar N e^{-\frac{x^2}{2a^2} - ikx} \left(\frac{-x}{a^2} - ik \right) dx$$

$$\# \quad p = \int_{-\infty}^{\infty} N e^{-\frac{x^2}{2a^2} + ikx} \cdot N e^{-\frac{x^2}{2a^2} - ikx} dx$$

$$1 = \int_{-\infty}^{\infty} N^2 e^{-x^2/a^2} dx = N^2 \int_{-\infty}^{\infty} e^{-x^2/a^2} dx = \boxed{a\sqrt{\pi}} N^2$$

$$1 = N^2 \left[\int_{-\infty}^{\infty} x e^{-x^2/a^2} dx + \frac{2}{a^2} \int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} dx \right]$$

$$\text{Take } u = -\frac{x^2}{a^2} \Rightarrow du = \frac{-2x}{a^2} dx$$

$$1 = N^2 \left[x e^{-x^2/a^2} + \frac{2}{a^2} \int e^u (-a^2 u) \frac{a^2 du}{-2\sqrt{-a^2 u}} \right]_{-\infty}^{\infty}$$

$$N = \frac{1}{\sqrt{a(\pi)^{1/4}}}$$

$$\langle x \rangle = \int_{-\infty}^{\infty} N^2 e^{-x^2/2a^2 + ikx} \cdot x \cdot e^{-x^2/2a^2 - ikx} dx = N^2 \int_{-\infty}^{\infty} x e^{-x^2/a^2} dx$$

$$\langle x \rangle = 0$$

$$\begin{aligned}
 \langle p_x \rangle &= i\hbar \int_{-\infty}^{\infty} N^2 e^{-x^2/a^2} \left(-\frac{x}{a^2} + ik \right) dx \\
 &= i\hbar N^2 \int_{-\infty}^{\infty} \left(\frac{-x}{a^2} e^{-x^2/a^2} + ik e^{-x^2/a^2} \right) dx \\
 &= i\hbar N^2 [0 + ika\sqrt{\pi}] \\
 &= +\hbar k N^2 a\sqrt{\pi} \\
 &= +\hbar k \frac{1}{a\sqrt{\pi}} \cdot a\sqrt{\pi} = +\hbar k
 \end{aligned}$$

Q. Find the probability that a particle wrapped in a box of width l can be found between $0.3l$ and $0.65l$ for the first excited state.

Sol.

$$\psi = \sqrt{\frac{2}{l}} \sin\left(\frac{n\pi x}{l}\right)$$

$$P = \int_{-\infty}^{\infty} \psi \psi^* dx$$

$$P = \int_{0.3l}^{0.65l} \sqrt{\frac{2}{l}} \sin\left(\frac{n\pi x}{l}\right) \sqrt{\frac{2}{l}} \sin\left(\frac{n\pi x}{l}\right) dx$$

$$P = \frac{2}{l} \int_{0.3l}^{0.65l} \sin^2\left(\frac{n\pi x}{l}\right) dx = \frac{2}{l} \int_{0.3l}^{0.65l} \frac{1 - \cos \frac{2n\pi x}{l}}{2} dx$$

$$P = \frac{1}{l} \int_{0.3l}^{0.65l} \left(1 - \cos \frac{2n\pi x}{l} \right) dx$$

$$P = \frac{1}{l} \left[x - \sin\left(\frac{2n\pi x}{l}\right) \cdot \frac{l}{2n\pi} \right]_{0.3l}^{0.65l}$$

$$P = \frac{1}{l} \left[0.65l - \frac{l}{2n\pi} \sin[2n\pi(0.65)] - 0.3l + \frac{l}{2n\pi} \sin[2n\pi(0.3)] \right]$$

$$P = \frac{1}{l} \left[0.35l - \frac{l^2}{4\pi^2} [\sin(2.6\pi) - \sin(1.2\pi)] \right]$$

$P =$

÷ Coordination number:

It is defined as the no. of nearest atoms which are directly surrounding a given atom.

Simple cubic : 8

Body centred : 8

Face centred : 12

Hexagonal closed Packing: 12

÷ No. of atoms per unit cell:

Simple cubic: $\frac{1 \times 8}{8} = 1$

Body centred: $\frac{1 \times 8}{8} + 1 = 2$

Face centred: $\frac{1 \times 8}{8} + \frac{1 \times 6}{2} = 4$

Hexagonal closed Packing: $\left(\frac{1}{6} \times 6 + 1 \times \frac{1}{2}\right) \times 2 + 3 = 6$

Primitive of side = a

Radius of atom = r

Simple cubic: $2r = a \Rightarrow r = \frac{a}{2}$



Body centred: $4r = \sqrt{3}a \rightarrow \text{diagonally}$
 $r = \frac{\sqrt{3}a}{4}$

Face centred: $4r = \sqrt{2}a$
 $r = \frac{a}{2\sqrt{2}}$

Hexagonal closed packing: $r = \frac{a}{2}$

Atomic Packing Fraction = $\frac{\text{Vol. of atoms per unit cell} = \frac{v}{V}}{\text{Vol. of unit cell}}$

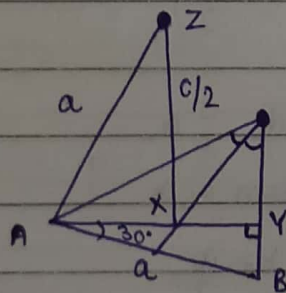
Simple cubic: $APF = \frac{\frac{4}{3} \pi r^3}{a^3} = \frac{4 \pi a^3 \cdot a^{-3}}{3 (2)^3} = \frac{2 \pi \cdot 1}{3 \cdot 4 \cdot 6} = \frac{\pi}{6} = 0.52$

Body centred: $APF = \frac{2 \times \frac{4}{3} \pi r^3}{a^3} = \frac{8 \pi \times \left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{8 \pi \times \frac{\sqrt{3}}{8}}{8} = \frac{\sqrt{3} \pi}{8}$
 $= 0.67$

Face centred: $APF = \frac{4 \times \frac{4}{3} \pi r^3}{a^3} = \frac{16 \pi \left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{16 \pi}{3 \times 8 \times 2\sqrt{2}} = \frac{\pi}{3\sqrt{2}}$
 $= 0.74$

Hexagonal closed packing: $APF = \frac{6 \times \frac{4}{3} \pi r^3}{V} = \frac{6 \times \frac{4}{3} \pi \left(\frac{a}{2}\right)^3}{V} = \frac{\pi a^3}{V}$

height $\rightarrow c$



$a = 2r$

$\cos 30^\circ = \frac{AY}{AB}$

$AY = AB \cos 30^\circ = \frac{\sqrt{3}a}{2}$

$AX = \frac{2}{3} AY = \frac{a}{\sqrt{3}}$

$a^2 = \frac{c^2}{4} + \frac{a^2}{3} \Rightarrow \frac{c^2}{4} = \frac{2a^2}{3} \Rightarrow c^2 = \frac{8a^2}{3}$

$V = \frac{6 \times \frac{4}{3} \pi r^3}{1} ; V = 6 \times \frac{1}{2} a \times \frac{a\sqrt{3}}{2} \times c$

$V = \frac{6 \times \frac{4}{3} \pi a^3}{8} = \frac{\pi a^3}{2} ; V = \frac{3a^2\sqrt{3}}{2} \times \frac{2\sqrt{2}a}{\sqrt{3}} = 3\sqrt{2}a^3$

$$APF = \frac{\pi a^3}{3\sqrt{2}a^3} = \frac{\pi}{3\sqrt{2}} = 0.74$$

Relation b/w density of crystal material
and lattice constant in a cubic lattice

Lattice constant = a ; density = ρ

No. of atoms per unit cell = n

Atomic wt. of material = M_A

Avogadro no. = N_A

Mass of material = ρa^3

Mass of material in terms of atomic weight = $\frac{M_A \times n}{N_A}$

$$\rho a^3 = \frac{M_A \times n}{N_A}$$

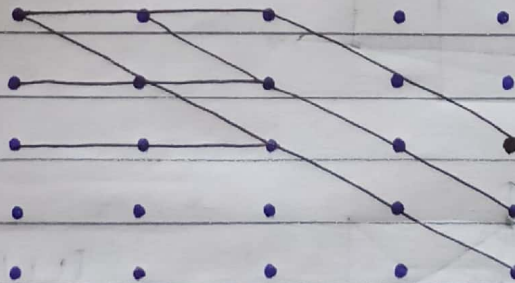
8. NaCl crystal has BCC structure. The density of NaCl is 2.18 g/cm^3 .
Calculate the distance b/w 2 adjacent atoms.

Solⁿ

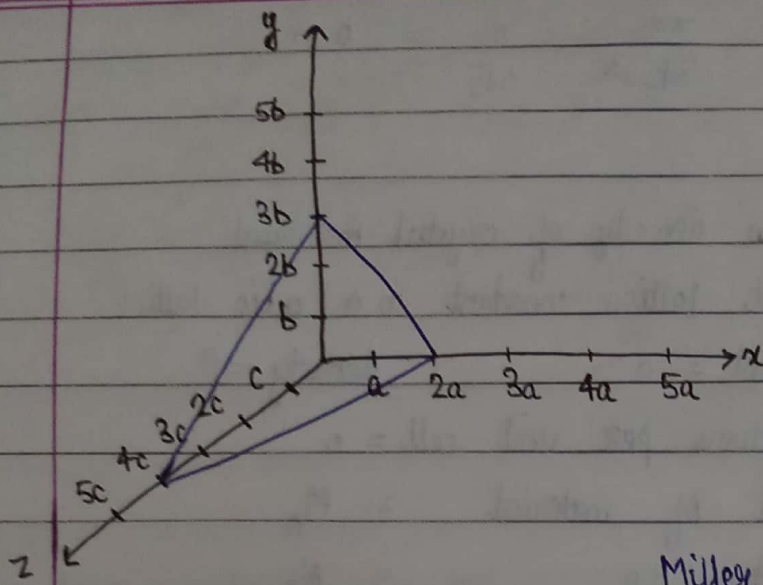
$$2.18 \times a^3 = \frac{58.5 \times 2}{6.022 \times 10^{23}}$$

$$a = \left(\frac{58.5 \times 2}{6.022 \times 10^{23} \times 2.18} \right)^{1/3}$$

Miller indices



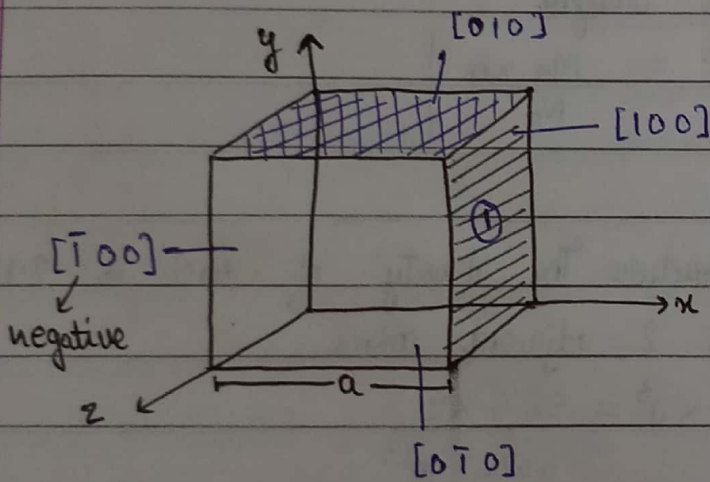
$[hkl] \rightarrow$ miller indices



1.	$2a$	$3b$	$4c$
2.	$\frac{2a}{a}$	$\frac{3b}{b}$	$\frac{4c}{c}$
3.	2	3	4
4.	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$
5.	$\frac{1 \times 12}{2}$	$\frac{1 \times 12}{3}$	$\frac{1 \times 12}{4}$

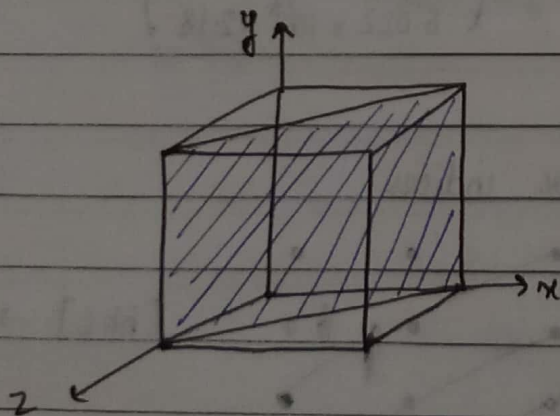
Miller indices
↓
[6 4 3]

→ 6 4 3

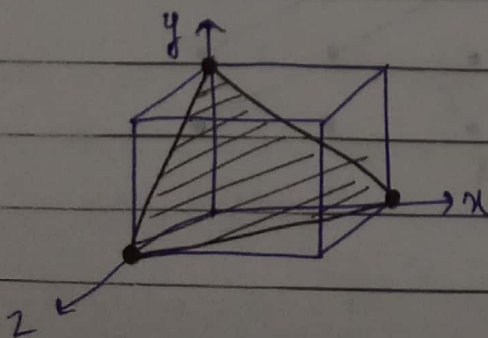


for plane ①,

a	∞	∞
1	∞	∞
1	$\frac{1}{\infty}$	$\frac{1}{\infty}$
1	0	0



[101]



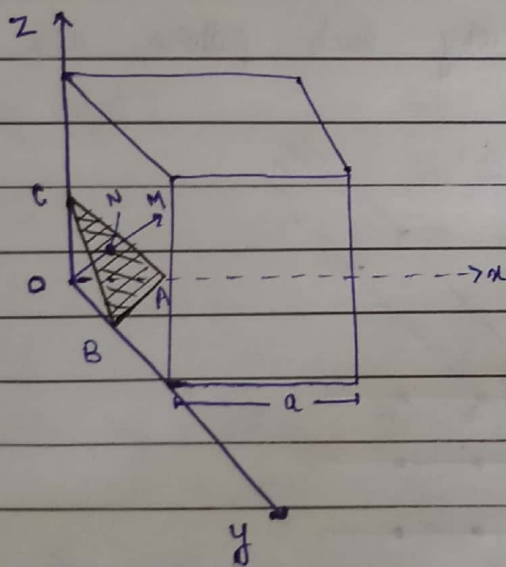
[111]

Eg.	6	10	15	
	$\frac{6}{30}$	$\frac{10}{30}$	$\frac{15}{30}$	
	$\frac{1}{5}$	$\frac{1}{3}$	$\frac{1}{2}$	
	5	3	2	
	5a	3b	2c	→ intercepts

Distance between the Miller planes (d)

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

a → lattice constant

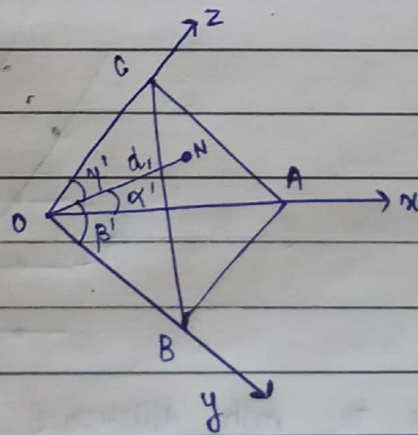


$$ON = d_1$$

$$OA = \frac{a}{h}$$

$$OB = \frac{a}{k}$$

$$OC = \frac{a}{l}$$



$$\cos \alpha' = \frac{d_1}{OA}$$

$$\cos \beta' = \frac{d_1}{OB}$$

$$\cos \gamma' = \frac{d_1}{OC}$$

$$\cos^2 \alpha' + \cos^2 \beta' + \cos^2 \gamma' = 1$$

$$\left(\frac{d_1}{OA}\right)^2 + \left(\frac{d_1}{OB}\right)^2 + \left(\frac{d_1}{OC}\right)^2 = 1$$

$$\left(\frac{d_1 h}{a}\right)^2 + \left(\frac{d_1 k}{a}\right)^2 + \left(\frac{d_1 l}{a}\right)^2 = 1$$

$$\frac{d_1^2}{a^2} (h^2 + k^2 + l^2) = 1 \Rightarrow d_1 = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$ON_1 = d_2$$

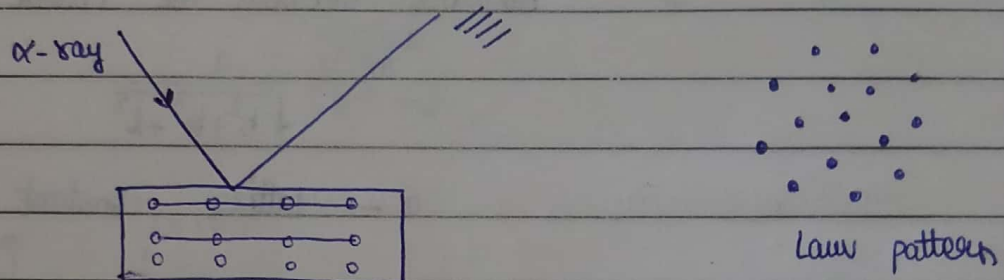
$$OA = \frac{2a}{b}, \quad OB = \frac{2b}{k}, \quad OC = \frac{2a}{l}$$

$$d_2 = \frac{2a}{\sqrt{b^2 + k^2 + l^2}}$$

$$d = d_2 - d_1 = \frac{a}{\sqrt{b^2 + k^2 + l^2}}$$

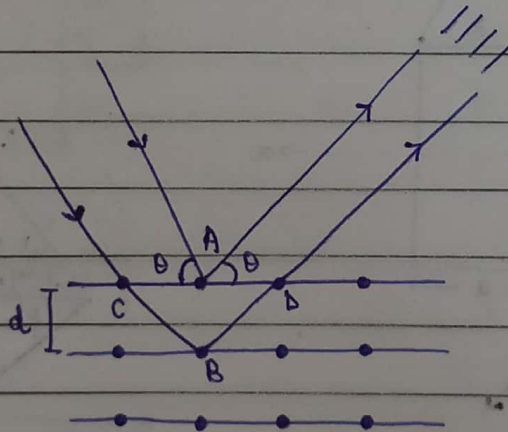
X-Ray diffraction

i) Laue method:



Laue couldn't explain as to why such pattern was formed.

ii) Bragg's Law:



Due to path difference b/w the 2 rays, we get maxima and minima.

$$\text{Path difference, } CB + BD = n\lambda$$

$$d \sin \theta + d \sin \theta = n\lambda$$

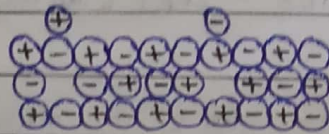
$$\boxed{2d \sin \theta = n\lambda}$$

$n \rightarrow$ order of spectrum

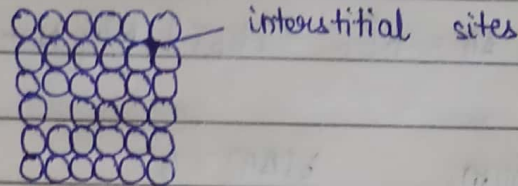
$d \rightarrow$ distance b/w the planes

Crystal defects

Schottky defect:



Frenkel defect:



The displaced ion adjusts itself into these interstitial sites.

Total no. of ways for shifting = $\underset{\substack{\downarrow \\ \text{for cation}}}{N_C} \times \underset{\substack{\downarrow \\ \text{for anion}}}{N_A} = (N_C)^2$

(i) Schottky defect:

$N \rightarrow$ total no. of atoms

$n \rightarrow$ no. of atoms shifted

$N_i \rightarrow$ no. of interstitial sites

$nA^+ \rightarrow$ cations

$nB^- \rightarrow$ anions

$\bar{E}_s \rightarrow$ avg. energy to generate 1 defect ; it is enthalpy

Average energy = $n\bar{E}_s$

No. of ways in which (W) = $[N_C]^2$
defects can be generated

$$W = \left[\frac{N!}{(N-n)! n!} \right]^2$$

Entropy (ΔS) = $R_B \log W$; $R_B \rightarrow$ Boltzmann constant

$$\Delta S = R_B \log \left[\frac{N!}{(N-n)! n!} \right]^2$$

Change in Helmholtz free energy (ΔA) = $n\bar{E}_s - T\Delta S$

$$\Delta A = n\bar{E}_s - T R_B \log \left[\frac{N!}{(N-n)! n!} \right]^2$$

$$\Delta A = n \bar{E}_s - 2k_B T \log \left[\frac{N!}{(N-n)! n!} \right]$$

$$= n \bar{E}_s - 2k_B T [\log N! - \log (N-n)! - \log (n!)]$$

Stirling approximation: $\log x! = x \log x - x$

$$\Delta A = n \bar{E}_s - 2k_B T [N \log N - N - (N-n) \log (N-n) + (N-n) - n \log n + n]$$

$$\Delta A = n \bar{E}_s - 2k_B T [N \log N - (N-n) \log (N-n) - n \log n]$$

at equilibrium, $\frac{\delta(\Delta A)}{\delta n} = 0$

$$\frac{\delta(\Delta A)}{\delta n} = \bar{E}_s - 2k_B T \left[\log(N-n) - \frac{n}{N-n} - \log n - 1 \right] + \frac{N}{N-n}$$

$$= \bar{E}_s - 2k_B T \left[\log \left(\frac{N-n}{n} \right) - \frac{N}{N-n} \right] + \frac{N}{N-n}$$

$$= \bar{E}_s - 2k_B T \left[\log \left(\frac{N-n}{n} \right) \right] = 0$$

$$\bar{E}_s = 2k_B T \log \left(\frac{N-n}{n} \right)$$

$$\frac{\bar{E}_s}{2k_B T} = \log \left(\frac{N-n}{n} \right)$$

Since $N \gg n$, $\exp \left[\frac{\bar{E}_s}{2k_B T} \right] = \frac{N}{n}$

$$n = N \exp \left[-\frac{\bar{E}_s}{2k_B T} \right]$$

(iii) Frenkel defect:

$$W = \frac{N!}{(N-n)! n!} \times \frac{N_i!}{(N_i-n)! n!}$$

$$\Delta S = k_B \log W = k_B \log \left[\frac{N!}{(N-n)! n!} \times \frac{N_i!}{(N_i-n)! n!} \right]$$

$$\Delta A = n \bar{E}_f - T \Delta S$$

$$\Delta A = n\bar{E}_f - T R_B \log \left[\frac{N!}{(N-n)! n!} \times \frac{N_i!}{(N_i-n)! n!} \right]$$

$$\frac{\delta(\Delta A)}{\delta n} = \bar{E}_f + R_B T$$

Stirling approximation:

$$\Delta A = n\bar{E}_f - R_B T [\log N! - \log (N-n)! - \log n! + \log N_i! - \log (N_i-n)! - \log n!]$$

$$\Delta A = n\bar{E}_f - R_B T [N \log N - N - (N-n) \log (N-n) + (N-n) - n \log n + n + N_i \log N_i - N_i - (N_i-n) \log (N_i-n) + (N_i-n) - n \log n + n]$$

$$\Delta A = n\bar{E}_f - R_B T [N \log N - (N-n) \log (N-n) - 2n \log n + N_i \log N_i - 2N_i - (N_i-n) \log (N_i-n)]$$

$$\frac{\delta(\Delta A)}{\delta n} = \bar{E}_f - R_B T \left[\cancel{N} + \cancel{(N-n)} \log (N-n) + \frac{N-n}{N-n} - 2 \log n - \frac{2n}{n} + \frac{N_i-n}{N_i-n} \right]$$

$$+ \log (N_i-n)$$

$$= \bar{E}_f - R_B T \left[\cancel{N} \neq \log \frac{N-n}{(N_i-n)^{-1}} - 2 \log n \right]$$

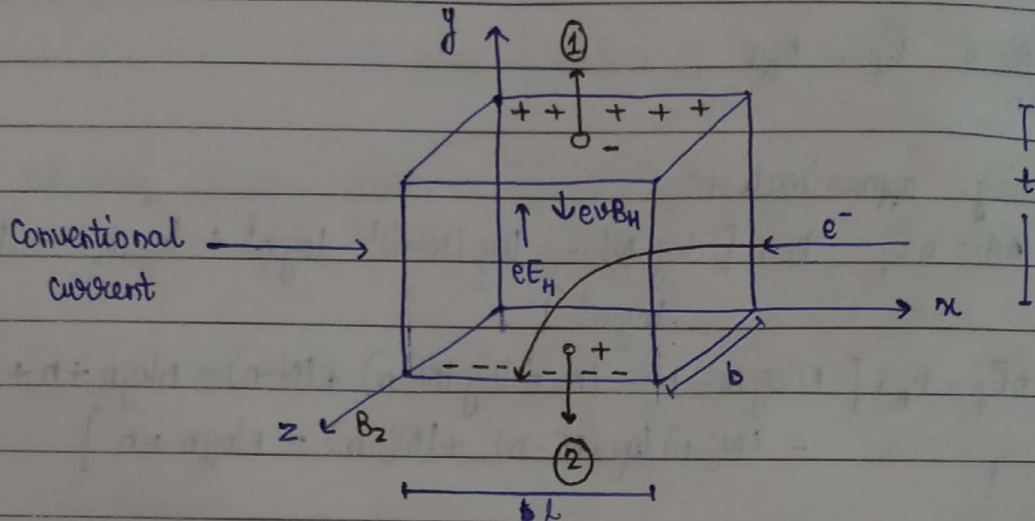
$$= \bar{E}_f - R_B T \left[\log \frac{(N-n)(N_i-n)}{n^2} \right]$$

$$\frac{\bar{E}_f}{R_B T} = \log \frac{(N-n)(N_i-n)}{n^2} \Rightarrow \exp \left[\frac{\bar{E}_f}{R_B T} \right] = \frac{NN_i}{n^2}$$

$$n = \sqrt{NN_i \exp \left[-\frac{\bar{E}_f}{RT} \right]}$$

UNIT-4

Hall Effect



$$E_H = \frac{V_H}{t}$$

$V_H \rightarrow$ Hall voltage

-ve charges will be collected on side 2.

+ve charges will be collected on side 1.

$$eE_H = evB_z \quad (\text{or } B_H)$$

$$v = \frac{E_H}{B_z}$$

$$I_x = nev = \frac{neE_H}{B_z}$$

$$I_x B_z = neE_H$$

$$R_H = \frac{1}{ne} = \frac{E_H}{I_x B_z}$$

$\rightarrow$ for e^-

$$R_H = \frac{1}{pe} = \frac{E_H}{I_x B_z}$$

$\rightarrow$ for holes

Current density, $J_x = \frac{I_x}{tb}$

$tb \rightarrow$ cross-sectional area

$$R_H = \frac{V_H \times tb}{I_x \times t \times B_z} = \frac{1}{ne}$$

$$R_H = \frac{1}{ne} = \frac{V_H b}{I_x B_z}$$

Applications:

- To check if specimen is p-type or n-type.
If R_H is -ve $\rightarrow$ n type
 R_H is +ve $\rightarrow$ p type
- To calculate carrier concentration (n/p)
- To measure the magnetic flux at any place using the Hall effect experiment

$$V_H = \frac{R_H I_x B_z}{b}$$

$$V_H \propto B_z$$

- To measure electromagnetic radiation
Poynting vector, $S = E \times B$
and $V_H \propto (E \times B)$

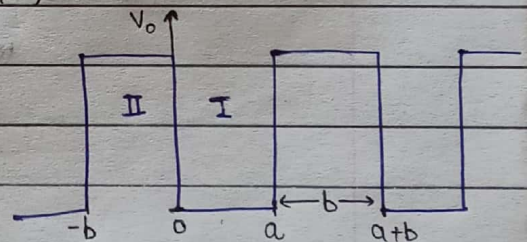
K. P. Model

(Kronig-Penney model)

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$$\text{I. } V_x = 0 \quad 0 < x < a$$

$$\text{II. } V_x = V_0 \quad -b < x < 0$$

Bloch Suggestion:

$$U_k(x) = U_k(x+a)$$

$$= U_k(x+2a)$$

(Periodic function)

$$1. \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} E \psi = 0 \quad \text{so,} \quad \frac{\partial^2 \psi}{\partial x^2} + \alpha^2 \psi = 0$$

$$2. \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V_0) \psi = 0 \quad \text{so,} \quad \frac{\partial^2 \psi}{\partial x^2} - \beta^2 \psi = 0$$

$$\text{Let } \alpha^2 = \frac{2m}{\hbar^2} E$$

$$\beta^2 = \frac{2m}{\hbar^2} (V_0 - E)$$

$$\psi(x) = U_k(x) e^{ikx}$$

$$\frac{\partial \psi}{\partial x} = e^{ikx} \frac{\partial U_k}{\partial x} + ik U_k(x) e^{ikx}$$

$$\frac{\partial^2 \psi}{\partial x^2} = e^{ikx} \frac{\partial^2 U_k}{\partial x^2} + ike^{ikx} \frac{\partial U_k}{\partial x} + ike^{ikx} \frac{\partial U_k}{\partial x} - k^2 U_k(x) e^{ikx}$$

$$\frac{\partial^2 \psi}{\partial x^2} = e^{ikx} \frac{\partial^2 U_k}{\partial x^2} + 2ike^{ikx} \frac{\partial U_k}{\partial x} - k^2 U_k e^{ikx}$$

So, 1. $\frac{\partial^2 U_1}{\partial x^2} + 2ik \frac{\partial U_1}{\partial x} + (\alpha^2 - k^2) U_1 = 0$

2. $\frac{\partial^2 U_2}{\partial x^2} + 2ik \frac{\partial U_2}{\partial x} - (\beta^2 + k^2) U_2 = 0$

for trial solution, $U_1 = e^{mx}$

$m \rightarrow$ arbitrary constant

$$\frac{\partial U_1}{\partial x} = m e^{mx}$$

$$\frac{\partial^2 U_1}{\partial x^2} = m^2 e^{mx}$$

1. $m^2 e^{mx} + 2ik m e^{mx} + (\alpha^2 - k^2) e^{mx} = 0$

$$(m^2 + 2ikm + \alpha^2 - k^2) e^{mx} = 0$$

$$m^2 + 2ikm + \alpha^2 - k^2 = 0$$

$$m = \frac{-2ik \pm \sqrt{-4k^2 - 4(\alpha^2 - k^2)}}{2}$$

$$2$$

$$m = \frac{-2ik \pm \sqrt{-4\alpha^2}}{2} = \frac{-2ik \pm 2\alpha i}{2}$$

$$2$$

$$m = -ik \pm i\alpha$$

$$m_1 = i\alpha - ik \quad ; \quad m_2 = -(i\alpha + ik)$$

$$m_1 = i(\alpha - k) \quad ; \quad m_2 = -i(\alpha + k)$$

We know, $U_1 = A e^{m_1 x} + B e^{m_2 x}$

$$U_1 = A e^{i(\alpha - k)x} + B e^{-i(\alpha + k)x}$$

2. also, $U_2 = C e^{(\beta - ik)x} + D e^{-(\beta + ik)x}$

Conditions:

1. $(U_1)_{x=0} = (U_2)_{x=0}$

3. $(U_1)_{x=a} = (U_2)_{x=-b}$

2. $\left(\frac{\partial U_1}{\partial x}\right)_{x=0} = \left(\frac{\partial U_2}{\partial x}\right)_{x=0}$

4. $\left(\frac{\partial U_1}{\partial x}\right)_{x=a} = \left(\frac{\partial U_2}{\partial x}\right)_{x=-b}$

$$U_1 (x=0) : U_1 = A + B$$

$$U_2 (x=0) : U_2 = C + D$$

$$U_1 (x=a) : U_1 = A e^{i(\alpha-k)a} + B e^{-i(\alpha+k)a}$$

$$U_2 (x=-b) : U_2 = C e^{-(\beta-ik)b} + D e^{(\beta+ik)b}$$

$$\frac{\delta U_1}{\delta x} = i(\alpha-k) A e^{i(\alpha-k)x} - i(\alpha+k) B e^{-i(\alpha+k)x}$$

$$\frac{\delta U_2}{\delta x} = (\beta-ik) C e^{(\beta-ik)x} - (\beta+ik) D e^{-(\beta+ik)x}$$

$$\left. \frac{\delta U_1}{\delta x} \right|_{x=0} = (i\alpha - ik) A - (i\alpha + ik) B \quad -①$$

$$\left. \frac{\delta U_2}{\delta x} \right|_{x=0} = C(\beta - ik) - D(\beta + ik) \quad -②$$

$$\left. \frac{\delta U_1}{\delta x} \right|_{x=a} = i(\alpha-k) A e^{i(\alpha-k)a} - i(\alpha+k) B e^{-i(\alpha+k)a} \quad -③$$

$$\left. \frac{\delta U_2}{\delta x} \right|_{x=-b} = (\beta-ik) C e^{-(\beta-ik)b} - (\beta+ik) D e^{(\beta+ik)b} \quad -④$$

from the coeff. of 4 eq's,

1	1	1	1
$i(\alpha-k)$	$-i(\alpha+k)$	$\beta-ik$	$-(\beta+ik)$
$e^{i(\alpha-k)a}$	$e^{-i(\alpha+k)a}$	$e^{-(\beta-ik)b}$	$e^{(\beta+ik)b}$
$i(\alpha-k)e^{i(\alpha-k)a}$	$-i(\alpha+k)e^{-i(\alpha+k)a}$	$(\beta-ik)e^{-(\beta-ik)b}$	$-(\beta+ik)e^{(\beta+ik)b}$

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sinh(\beta b) \sin(\alpha a) + \cosh(\beta b) \cos(\alpha a) = \cos k(a+b)$$

$$\alpha^2 = \frac{2mE}{\hbar^2}$$

$$\beta^2 = \frac{2m(V_0 - E)}{\hbar^2}$$

$$(\beta^2 - \alpha^2) = \frac{2m}{\hbar^2} (V_0 - 2E)$$

$$V_0 > E$$

$$(p^2 - \alpha^2) = \frac{2m}{\hbar^2} V_0$$

$$\frac{mV_0}{\hbar^2 \alpha \beta} \sinh(\beta b) \sin(\alpha a) + \cosh(\beta b) \cos(\alpha a) = \cos k(a+b)$$

$$V_0 \rightarrow \infty$$

$$V_0 b \rightarrow \text{finite}$$

$$b \rightarrow 0$$



$$\sinh(\beta b) = \beta b$$

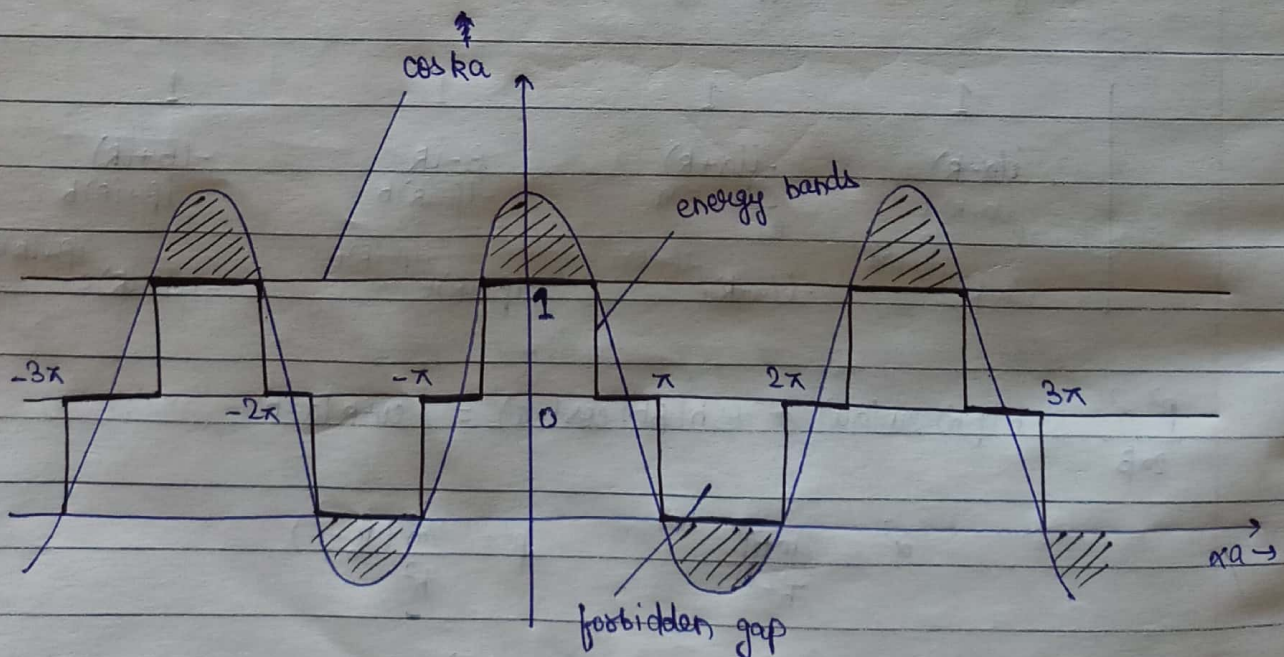
$$\cosh(\beta b) = 1$$

$$\frac{mV_0}{\hbar^2 \alpha \beta} \times \beta b \sin(\alpha a) + \cos(\alpha a) = \cos ka$$

$$\text{let } P = \frac{mV_0 b a}{\hbar^2}$$

$$\frac{P \sinh(\alpha a)}{\alpha a} + \cos \alpha a = \cos ka$$

$$\text{let } P = \frac{mV_0 b a}{\hbar^2} = \frac{3\pi}{2}$$



Shaded regions don't exist.

Case 1: $P \rightarrow \infty$

$$b \rightarrow 0$$

$$\sin(\alpha a) = 0$$

$$\alpha a = \pm n\pi$$

$$\frac{\sqrt{2mE}}{\hbar} \times 2\pi = \alpha = \pm \frac{n\pi}{a}$$

$$E = \frac{n^2 \hbar^2}{8ma^2}$$

Case 2: $P \rightarrow 0$

$$\cos(\alpha a) = \cos(ka)$$

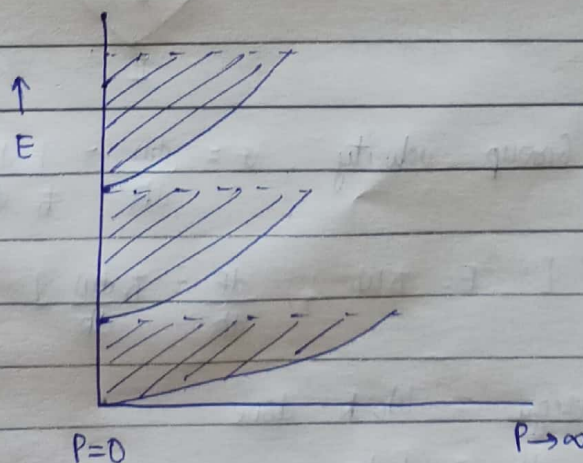
$$\alpha a = ka$$

$$\boxed{\alpha = k}$$

$$\frac{2mE}{\hbar^2} \times 4\pi^2 = \left(\frac{2\pi}{\lambda}\right)^2$$

$$E = \frac{p^2}{2m} \quad ; \quad \lambda = \frac{h}{p}$$

$$\boxed{E = \frac{1}{2} mv^2}$$



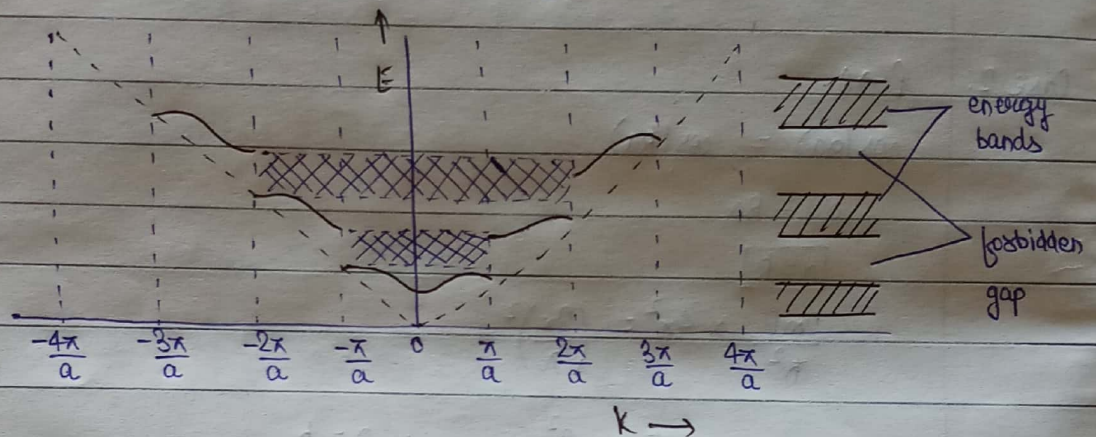
E-K Diagram
 energy $\downarrow$ wave no. $\rightarrow$

$$E = \frac{\hbar^2 k^2}{8ma^2}, \quad k = \frac{2\pi}{\lambda} = \frac{n\pi}{a \sin \theta} = \pm \frac{n\pi}{a}$$

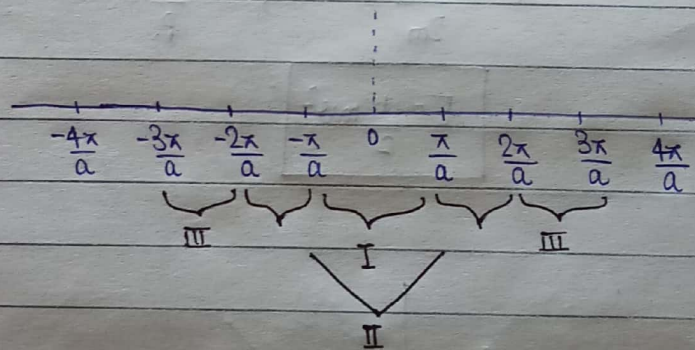
Scattering, $2a \sin \theta = n\lambda$

$$\frac{2}{\lambda} = \frac{n}{a \sin \theta}$$

for linear variation, $\theta = \pm \frac{\pi}{2}$



BRILLOUIN ZONES:



Effective mass: Group velocity, $v = \frac{d\omega}{dk} = \frac{1}{\hbar} \frac{dE}{dk}$

$$[\because E = \hbar \omega; \quad \frac{dE}{dk} = \hbar \frac{d\omega}{dk}]$$

Increase in energy = Work done

$$dE = F \cdot dx$$

$$dE = e \mathcal{E}_1 dx = e \mathcal{E}_1 v dt$$

($\mathcal{E}_1 \rightarrow$ el. field)

$$\frac{dE}{dk} \times dk = e \mathcal{E}_1 \times \frac{1}{\hbar} \frac{dE}{dk} \times dt$$

$$\frac{dk}{dt} = \frac{1}{\hbar} e \mathcal{E}_1$$

$$\hbar \frac{dk}{dt} = e \mathcal{E}_1$$

$$F = e \mathcal{E}_1 = \frac{dp}{dt}$$

$$a = \frac{dv}{dt}$$

$$a = \frac{d}{dt} \left[\frac{1}{\hbar} \frac{dE}{dk} \right]$$

$$a = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \times \frac{dk}{dt}$$

$$a = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2} \times e \mathcal{E}_1$$

$$m^* = \hbar^2 \frac{d^2 E}{dk^2}$$